

Automated Pill Dispenser

Senior Project Final Report

Design Team 05

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Abstract

The Automated Pill Dispenser is a complete solution to reminding, organizing, and maintaining prescription pills. The device remembers a user's schedule in order to remind them when it is time to take their dosage. When it is time for the user to take a dose, the machine will verify the correct user via pin code and dispense the correct variety of pills. The pills are stored in secure compartments that can be accessed by the user at any time. The smart pill dispenser is Wi-Fi enabled to share usage statistics with family members or medical professionals. The device has a battery backup so a patient may continue to dispense medication in a power outage.

1. Problem Statement

1.1. Need

Prescription medication can be challenging to manage for people. When dealing with medication, people must manage their prescriptions, remember to take their medication on time, and handle the daily dosages. In 2021, 65% of adults over the age of 18 took some sort of prescription medication during the past 12 months. Currently, medication containers can be easy to forget about, hard to open and lack any indication of usage for loved ones or medical professionals. More advanced and fully automated pill dispensers are unnecessarily expensive, making them not accessible to most of the public. People using prescription medication need a way to organize and take their medication accurately to limit stress for themselves and their loved ones.

Authors: RL

1.2. Objective

The objective of this project is to design and prototype a device to dispense a quantity of pills in a configurable way. This device will contain features to assist both caregivers and medication takers with medication management needs. The user will provide the system with a configuration of what pills to dispense for a given time. It will contain a setting that turns on or off a sound to alert the user to when they need to take their pills. The device will also have an option to dispense the pills for any time of day on demand to fill up pill boxes. The device will use a feeding mechanism to properly allot pills for the desired time slot and dispense them out of central chute. This feeding

mechanism will contain software checking to ensure the proper number of pills have been dispensed.

Authors: CB, RO

1.3. Background

In 2021, 64.8% of adults took some sort of prescription medication during the past 12 months (Cohen & Mykyta, 2021). Since over half of the population reported taking medication in 2021, there is a demand for organization systems to aid with medication management. Automatic pill dispensers are innovative devices used to improve how people manage their prescriptions. Medication organization systems can range from single-dosage plastic containers to fully automated dispensers. In a 2023 study of patients with hypertension, 46.5% of participants reported moving their medication from the original pill bottles to some sort of pill dispenser, sorter, or other container (Genelin et al, 2023). When managing medications, it is useful to have some sort of organization system to help track quantity and ensure correct dosage of the medication.

The basic concept behind automatic pill dispensers is to automate parts of the medication management process that can be difficult and/or tedious for medication users. For example, the 2023 study on pill management techniques from hypertension patients found that patients who combined prescriptions into one pill bottle often underestimated the pill count (Genelin et al, 2023). This type of uncertainty surrounding how many pills remain after combining pill bottles can cause people to forget to fill their prescriptions at the correct times. An automatic pill dispenser could have a feature that keeps track of how many pills of each prescription remain and then alert the user when they need to refill. Managing multiple prescriptions is time-consuming for the medication user or their caretaker. Some people are unable to manage their medications independently, so having an organization system to assist both themselves and their caretakers can save time and reduce dependence on others when taking their medications. Features like scheduled dispenses, medication tracking and organization, alerts for dispenses and refills, and pill verification can all aid in streamlining the medication management process while ensuring users can take their prescriptions correctly.

Authors: RL

Currently, pill dispensers are mostly manually filled organizers or expensive automated systems. Many inexpensive pill organizers consist of plastic compartments labeled by time or the day of the week. While these are simple and easy to use, they aren't practical, they rely solely on the user's memory and don't provide any features such as alerts or tracking. Automated pill dispensers include built in alarms to notify users to take their medication, along with sorting, tracking and dispensing medications at specific times. The issue with these automated dispensers is the cost, causing many patients to miss out on accessibility and easy way of life.

Existing automated pill dispensers utilize mechanisms such as rotating trays, vibrating plates, or spiral chutes to dispense medication at desired times. These systems help users with auditory or visual alerts, they often struggle with accurately dispensing pills of varying sizes and shapes (Minaam, 2018). Along with this many dispensers rely on pre-filled medication packets, which can limit their use for those who take multiple medications from different sources. Incorporating sensors to verify correct number of pills has also been implemented, reducing the number of overdoses or missed doses, these systems are very expensive and require complex calibration to work effectively, which also limit their use for those without technical knowledge (Vaish, 2024).

Manufacturers also introduced smart pill dispensers that allow internet connectivity and AI to enhance medication management (Minaam, 2018). Some models include biometric security features to prevent unauthorized access, while others send notifications via a Mobile Application to caregivers when a dose is missed. These systems often require a monthly subscription or complex set ups that are not user friendly for the elderly or those with certain disabilities.

A few products are in the market that provide pill dispensing option to patients namely, Pointells Automatic Pill Dispenser, e-Pill Voice Pro, Hero smart pill dispenser, and MedaCube (Clark). The Hero smart pill dispenser is one of the more popular products used Globally, dispensing only one type of pill it sorts and dispenses user's dose. Adding the medication list into the app to receive pill-time reminders, missed-dose alerts, and track what users took and when.

Author: JP

There are existing patents that relate to automated pill dispensing. These patents use various techniques for dispensing pills and automating the process. A patent from

David W. Baarman is a stackable pill dispensing system with a user interface. The UI provides the user with information on pill inventory, and a pill taking schedule. Multiple pills dispensing packages are stacked on top of the pill dispensing base. Each pill dispensing package has its own dispensing element, which is chosen based on what is most effective for the pill shape. Pill dispensing elements come in different designs that allow the system to work with various pill sizes and shapes. To dispense a single pill, a pill dispensing package's dispensing element is actuated. In automated mode, the control system is used to keep track of a user's pill taking schedule. The control system communicates with each pill dispensing package to coordinate a multi-pill dispense. This patent also provides multiple potential ways of achieving security to ensure pills are only dispensed to the user it's designed for. Solutions mentioned include biometric scanning, passwords, a voice identification system, and a proximity sensor to determine if the dedicated user is close to the machine. Loading the pills into this system is meant for a pharmacy or dispensing facility. (Baarman, 2015)

Another patent from Michel Poirier takes a separate approach in the automation of pill dispensing. While this system also dispenses a configurable set of pills, it aims to fill pill boxes as well. The device is also capable of dispensing single doses at a time. This patent uses a spring-loaded cartridge to hold pills. Cartridges have different adapters for support a variety of pill shapes and sizes. Multiple cartridges sit on the dispenser base in a circular fashion. To dispense a pill in this system, the circular magazine is rotated to the dispensing position. In this dispensing position pills are ejected from the magazine and provided to the user. The device is intended for use in various environments, such as a patient's home, pharmacies, and nursing homes. An electronic system allows caregivers to monitor pill distribution wirelessly. (Poirier, 2022)

The patents discussed focus on portability, usability, and modularity. They also provide the option for caretakers to monitor a patient's activity.

Author: RO

Current designs suffer from a few key limitations that make them less practical for the average user. Any given design will often compromise on either usability or cost. Some current designs use a system akin to that of the first patent mentioned, where the pharmacy gives customized packages to the patient for use in their device. These packages are either containers of a single pill to allow easier dispensing or they are dosed

bags that are given to the user on a schedule. These devices are generally less flexible and require a high amount of coordination between the patient and their pharmacy to get the correct packages for dispensing in their machine.

Another common set of designs that falls into this category are smart pill boxes. Smart pill boxes generally have a small display for setting medication dispensing schedules. These devices are small and portable but require the user to input their pills into the device in the correct dosages and open the lids of the compartments on a schedule (Hartung, 2022). These devices are very affordable but suffer usability issues for those with movement disabilities who can have trouble filling up pill boxes with their daily doses of medication.

Other automated pill dispensers are built to accept any pills. This class of designs takes in sorted pills and are set up to dispense them on a schedule via a phone or computer application (Hartung, 2022). These designs also have more requirements for their functionality. They must use one mechanism to reliably and repeatably dispense pills of different sizes. Due to the difficult nature of the problem, many designs become complex and large. With this complexity their price can often increase rapidly. These products can cost over 1000 dollars paid upfront or rented for a subscription that can cost over 300 dollars each year.

Our Design will share many similarities with ideas that have already been prevalent in current design and research. Similar to current designs, our design will take in sorted pills and combine them together at the time they are needed to create the correct dosage for the user. It will also provide a similar alarm and missed dosage functionality to other comparable products on the market (Savvides, 2024). There will also be functionality to dispense dosages early on request from the user (Hartung, 2022). Similar to a paper by Anisha Vaish, our approach to dispensing pills will need to be adjusted based on the size and shape of the pill to ensure smooth operation. The design will have an emphasis on simplicity to make it usable for anybody. In contrast to other current designs, this innovative design is much simpler and cheaper to produce. With the prevalence of 3d printing as an available public resource in libraries and maker spaces, the design will be simple enough to build independently. The hope is that a semi-skilled hobbyist maker could buy or build a similar machine.

Author: CB

Automatic pill dispensers have improved many lives by enhancing accuracy and reducing the burden on users and caregivers. Current designs offer a lot of valuable features such as tracking, alerts and security. However, these designs are limited by their costs, complex setups and limited flexibility with versatile pill shape and size. This highlights the need for a more accessible, cost effective and user-friendly solution. Our project aims to build upon previous research and existing products to provide an innovative solution that can aid in medication management for those in need.

Author: JP

1.4. Marketing Requirements

The system should:

- Help users remember to take their pills on time.
- Manage the dispensing and organization of each dose.
- Store pills safely in separate compartments.
- Alert when pills need to be refilled.
- Lock to securely store pills, ensuring convenience and safety for the user.
- Be small enough to fit on top of kitchen or bathroom counters.

Authors: RL, JP

2. Engineering Analysis

2.1 Design Solutions

2.1.1 Circuits and Electronics

The system will have two sources of power, a 6 VDC wall adapter plug, and a backup battery system. This is to ensure that if there is power lost from the wall adapter plug that the device can still be operational off a backup battery. The system will use a single pole double throw relay to switch between the two power sources. There will be three power rails for the components of the circuit. 6 V will be used for the servos, 12 V for the display backlight, and 3.3 V for the rest of the circuit. The 6 V will be pulled off of either the wall adapter or the battery supply, the 12 V will be doubled from the 6 V and the 3.3 V will be stepped down to with a linear regulator. The design will also include filtering capacitors to help smooth out any voltage fluctuations. There will also be a mechanical master switch to turn the device on and off.

The overall circuit design is multiple input and output devices connected to the central microcontroller unit (MCU). The output devices include the 2.4 inch TFT LCD display and Wi-Fi module. The input devices include a push button and an encoder to control the screen. For security purposes, the circuit uses a pincode to confirm the identity of the user. There are also internal LEDs, phototransistors and lasers for pill dispense verification and quantity within the container.

For the printed circuit board layout, there will be two boards in the device. The power board will have the relay circuit, 6 V to 3.3 V voltage regulator, power filtering and servo connections. The MCU board will have 3.3 V rail, device connections to sensors and display, signal output to servos, Wi-Fi module and microcontroller connections.

Author: RL

2.1.2 Communications

The communications standards to be used between the different elements of our device are SPI and UART. This decision is due to our familiarity with the standards and their widespread adoption across the industry for communicating between different

components. For communications between the device and other computers, Wi-Fi will be implemented due to its higher convenience over a physical cable like ethernet. Most houses do not have networking cables running throughout the home and even less have networking cables running into the kitchen.

Author: CB

2.1.3 Electromechanics

The servos will be controlled with PWM signals from the microcontroller. The change in PWM controls the angle that the motor rotates at. The servos will provide precise motion control focusing on position and feedback. The servo motor system provides an electric motor (DC, AC), A feedback sensor (a potentiometer that measures position, speed or angle), Controller (Compares actual position to desired one and adjusts input signal), and Mechanical linkage (Converts motor rotation into required mechanical movement).

Author: RO

2.1.4 Embedded Systems

The selected MCU is STM32H5. The MCU supports the communication standards that are needed like SPI and UART. The microcontroller needs to support several SPI interfaces that are needed to interact with the Wi-Fi coprocessor and the screen.

2.1.5 Computer Networks

The device needs to be able to receive internet packets on the TCP/IP protocol on a wireless standard. The current solution is to use a separate Wi-Fi enabled chip that can be controlled from the main MCU. This Wi-Fi chip can be controlled through SPI to receive and transmit messages

Author: CB

2.1.6 Controls

The system control consists of using a servo to spin the wheel to pick pills verify the pick using sensors. Phototransistors will be used to read values and verify the presence of a pill. The motors are then triggered to open the gate allowing the pills to advance.

Author: RO

3. Engineering Requirements Specification –

Table 3.1 *Engineering Requirements for Marketing Requirements*

Marketing Requirements	Engineering Requirement	Justification
2, 5	Automatically dispense one pill every 5 seconds.	The dispensing time needs to be fast enough to make it convenient for the user, while still ensuring accuracy of the pills dispensed.
7	The device will be no taller than 18 inches to fit under normal height kitchen cabinets.	The device should be convenient for a user to fit in a standard kitchen.
1	Alert the user when a dosage is to be taken.	Alerts are crucial for automating the pill taking system. This helps keep the user on schedule with their medication.
1, 2	The device will have a configurable schedule for dispensing dosages.	Users will need to configure their device with their own dosage schedule. This allows users to customize their pill dispenser to fit their needs.
6	Stop unauthorized users from tampering with the pills.	Secures pills so that unintended people do not take them.
4	Determine when the internal pill containers are under a configurable quantity and alert the user that they need to be refilled.	This prevents a user from running out of pills mid dispense and only receiving a portion of their scheduled dosage. Also, it can serve as a reminder to fill prescriptions.
2, 3	Support storage and dispensing from 4 distinct pill channels.	Allows flexibility by allowing up to 4 different pill types in a dosage.

		Most people have multiple prescriptions so this will allow users to organize most of their medications.
2	Deliver the correct number of pills to the end user at least 90% of the time.	Ensure a high standard of accuracy to ensure the safety of the client.

Marketing Requirements

1. Help users remember to take their pills on time.
2. Manage the dispensing and organization of each dose accurately.
3. Store pills safely in separate compartments.
4. Alert when pills need to be refilled.
5. Offer the ability to fill external pill boxes for travel.
6. Lock to securely store pills, ensuring convenience and safety for the user.
7. Be small enough to fit on top of kitchen or bathroom counters.

4. Engineering Standards Specification –

SAFETY

This project will implement standards to ensure consumer safety and reliability. Data formats and design methods will be followed in a way that allows others to understand how the product operates.

The UL Anti-theft device testing and certification evaluates security of gun safes, lock boxes, and other devices meant for security. The pill dispenser needs to be theft deterrent to avoid pills being easily taken. The pill device will have a locking mechanism to prevent unverified users from accessing and tampering with medication within the device. While the device will not get certification with this standard, it will closely follow its guidelines to appropriately deter theft.

IEC 60038 defines standardized nominal voltages used in AC and DC to ensure compatibility and safety. It specifies nominal voltage values for low, medium, and high voltage systems. AC systems are as follows: Extra Low-Voltage (<50V), Low Voltage (50-1000V), and High Voltage (>1000). DC systems are as follows: Extra Low-Voltage (<120V), Low Voltage (120-1500V), and High Voltage (>1500).

IEC 60059 defines a set of preferred nominal current ratings for electrical devices and equipment. Its purpose is to ensure uniformity, safety, and interchangeability. The current ratings are listed below.

Standard current ratings A									
1	1,25	1,6	2	2,5	3,15	4	5	6,3	8
10	12,5	16	20	25	31,5	40	50	63	80
100	125	160	200	250	315	400	500	630	800
1000	1 250	1 600	2 000	2 500	3 150	4 000	5 000	6 300	8 000
10000	12 500	16 000	20 000	25 000	31 500	40 000	50 000	63 000	80 000
100000	125 000	160 000	200 000						

Figure 4.1 IEC 60059 standard current ratings

COMMUNICATION

ASCII is a character encoding standard that uses 7-bit character codes to represent the English alphabet, numbers, and control characters. This project will implement this form of character encoding because it is very commonly used among all electronic devices. ASCII will be used when communicating with peripherals such as the LCD screen, and data logs will be stored in an ASCII format.

ISO/IEC 21778:2017 is a standard that defines the appropriate syntax for a file to be considered valid JSON syntax. JSON is a commonly used format for data-exchange because of its readability and widespread support for parsing in common programming languages. This project will implement JSON to package log data and send to clients.

UART and SPI are serial communication protocols that allow a microcontroller to communicate with peripherals. The pill dispenser will use electronic devices such as an LCD screen, and Wi-Fi chip that all use communication through these protocols.

IEEE 802.11 is a standard that specifies protocols to enable wireless local area network communication. This standard is the basis for Wi-Fi communication, which will be used by the pill dispenser to transmit information about device usage.

DESIGN

ISO/IEC 14882:2024(E) is a C++ standard that defines syntax and features of the programming language. It is a set of rules for compilers and standard libraries to follow. The pill dispenser project will be compiled with tools that follow this standard.

5. Accepted Technical Design

5.1. Hardware Design:



Figure 5.1.1 Block Diagram of Pill Dispenser's Level 0 Functions

Table 5.1.1 Level 0 Functional Requirements

Module	Pill Dispenser
Designer	Connor Beaven, Rachelle LaBrie, Ryan Oderkirk, Josue Panchana
Inputs	Pill Schedule: Configuration info that describes dates/times that pills should be dispensed, and which pills should be dispensed. Pill dispense configuration: Configuration info that describes which feeder holds what type of pills. User input for dispensing: UI that the user will interact with to tell the machine to dispense pills.
Outputs	Pill reminder alert: Alarm that will remind the user to dispense their pills Dispensed pills: Pills to be dispensed in accordance with the user's configuration User interface: Provide feedback to the user on dispensing operation.

Level 0:

The device will store pills and dispense them with a configuration that allows the user to describe the quantity of pills along with the frequency of their dispensing. The device will notify the user when it is time to take their pills through the user interface. The user can then initiate the dispense through a button input.

Authors: CB, RO

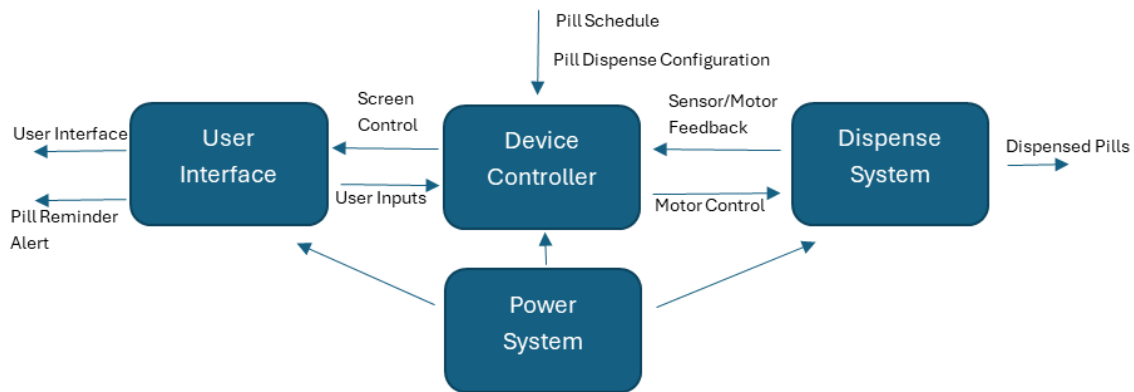


Figure 5.1.2 Level 1 Functional Block Diagram

Table 5.1.2 User Interface Inputs, Outputs, and Functionality

Module	User Interface
Inputs	User input, power, device controller inputs
Outputs	Device controller, user alerts
Functionality	The user interface is the connection from the user to the rest of the system. This is how the system will receive configurable inputs and communicate outputs to the user.

Table 5.1.3 Power System Inputs, Outputs, and Functionality

Module	Power System
Inputs	120 V from outlet
Outputs	Power to interface, controller and dispense system
Functionality	This module will provide power to the rest of the system

Table 5.1.4 Device Controller Inputs, Outputs, and Functionality

Module	Device Controller
Inputs	User input, power, device controller inputs, sensor readings, client data requests
Outputs	Device controller, user alerts, data responses

Functionality	The user interface is the connection from the user to the rest of the system. This is how the system will receive configurable inputs and communicate outputs to the user.
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Table 5.1.5 *Dispense System Inputs, Outputs, and Functionality*

Module	Dispense System
Inputs	Power, device controller
Outputs	Device controller
Functionality	The dispensing system will be the final stage to get the correct pill dosage to the user by storing, verifying and dispensing pills.

Level 1:

The level 1 functional block diagram, specified in Figure 5.1.2, is broken down into the user interface, the device controller, the dispense system, and the power system. The user interface block consists of the. Physical inputs and output that allow the user to interact with the device. This will be implemented through button inputs and a screen output. The device controller block coordinates all inputs and outputs in the system. It is responsible for reading UI inputs, writing UI screen outputs, controlling the servos, reading from sensors, and reading/responding to data requests. The dispense system block contains the mechanical parts of the device, such as motors and sensors, that will physically dispense and verify a dispense. The power system block is responsible for providing sufficient power to the rest of the system.

Authors: RO

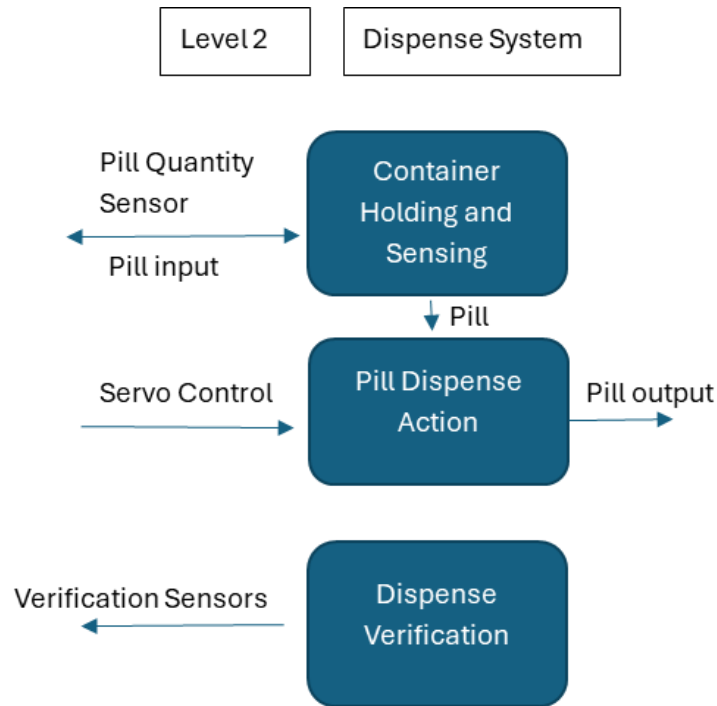


Figure 5.1.3 Level 2 Functional Diagram of Dispense System

Table 5.1.6 Container Holding and Sensing Inputs, Outputs, and Functionality

Module	Container Holding and Sensing
Inputs	Quantity of low pills from device controller, sensor power from power system
Outputs	Sensor reading to device controller, pills from container storage
Functionality	The container stores the different pills and senses how full each container is.

Table 5.1.7 Pill Dispense Action Inputs, Outputs, and Functionality

Module	Pill Dispense Action
Inputs	Pills, servo control from device controller
Outputs	Pills
Functionality	Dispense the pill

Table 5.1.8 Pill Verification Inputs, Outputs, and Functionality

Module	Pill Verification
--------	-------------------

Inputs	Pill from dispense action
Outputs	Sensor reading to device
Functionality	The pill verification

Dispense System Level 2:

The level 2 functional diagram of the dispense system (Figure 5.1.3) includes three different modules: Container Holding and Sensing, Pill Dispense Action, and Pill Verification. The Container Holding and Sensing module (Table 5.1.6) stores the pills until the dispense cycle starts. There is also a quantity sensor that outputs information about the total number of pills in the container. Once the dispense cycle begins, the container outputs pills to the Pill Dispense Action module. The Pill Dispense Action module (Table 5.1.7) represents the process of configured dispenses. The module inputs pills and commands to control the servos. The output is the correct number of pills being dispensed. The last module is the Pill Verification module, whose functionality is shown in Table 5.1.8. This module outputs verification information to ensure the correct dosage is dispensed to the user.

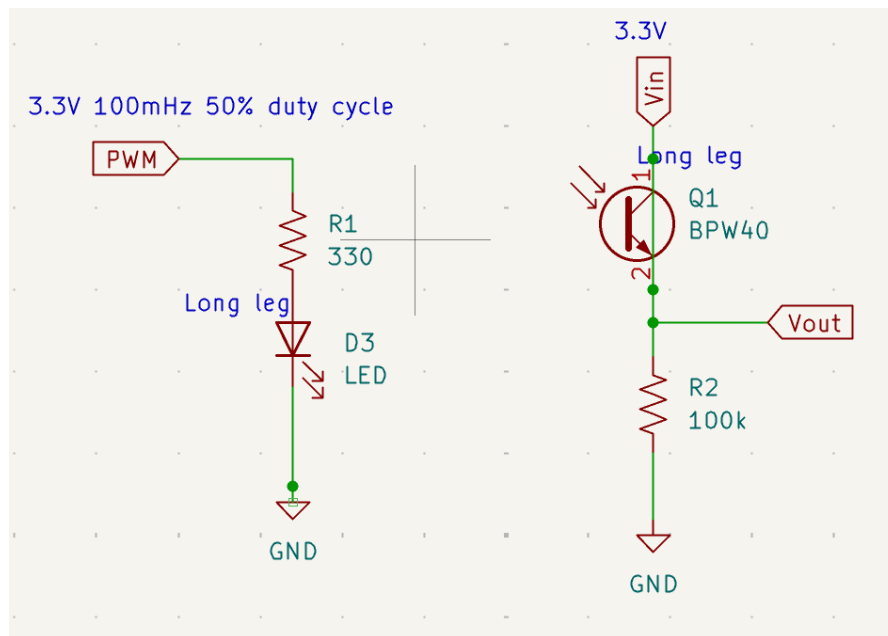


Figure 5.1.4 Phototransistor and LED schematic

To implement the Pill Verification module with hardware, the pill will pass through the dispense shoot and its presence will be detected to verify a successful dispense attempt. On opposite sides of the shoot, there will be a phototransistor and an LED. The pill will cover the light from the LED, which will be sensed by the phototransistor and read by the microcontroller.

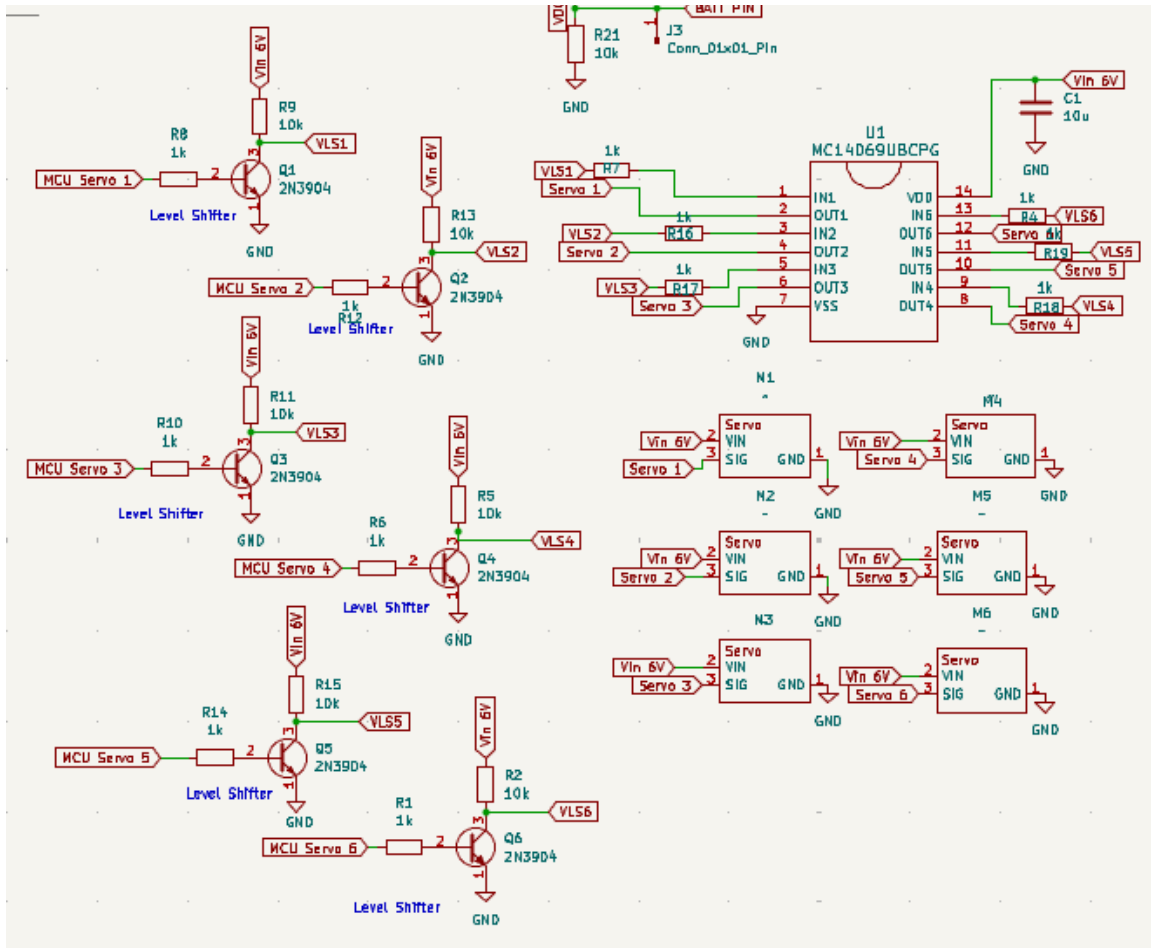


Figure 5.1.5 *Level Shifters and Inverter for Servo Control Circuit Schematic*

The servo control circuit supports the connection of up to six total servos. To send the correct amplitude of PWM to control the servos, the PWM signals from the MCU need to be shifted to 6 V. The level shifter inverts the PWM signal so the circuit also has an inverter to invert the PWM signal. The circuit schematic in Figure 5.1.5 shows the connections of the level shifts to the inverter chip and then to the servo motors.

Author: RL

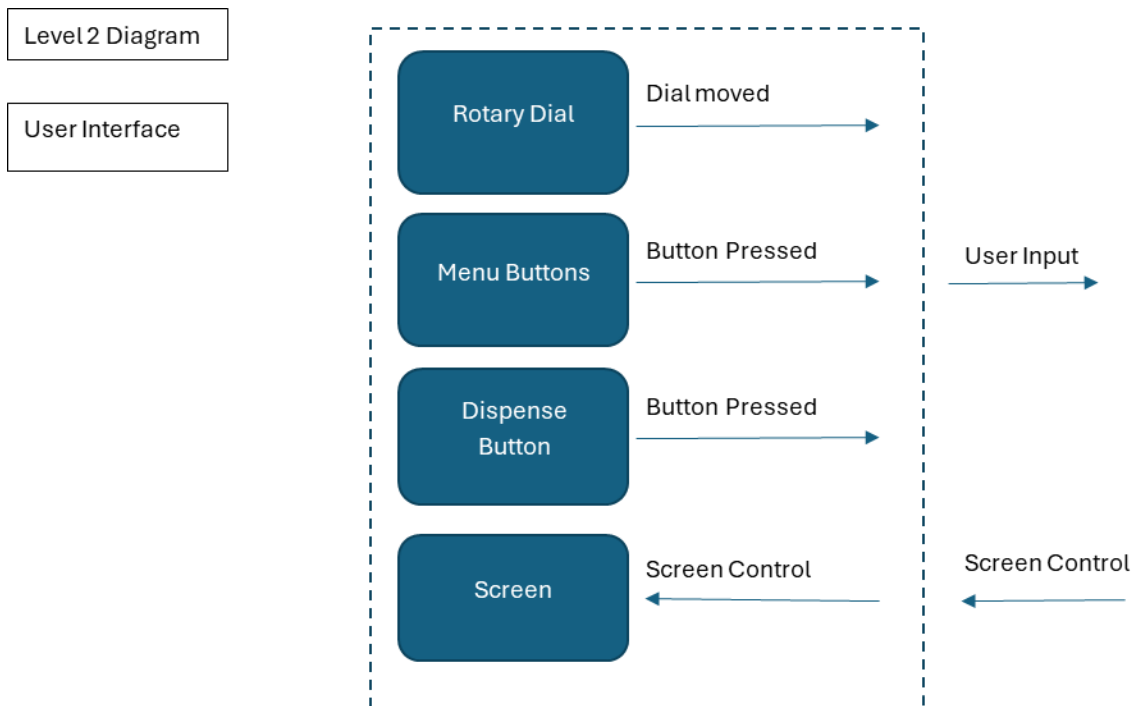


Figure 5.1.6 Level 2 Functional Diagram of User Interface

Table 5.1.9 Rotary Dial Inputs, Outputs, and Functionality

Module	Rotary Dial
Inputs	Mechanical Rotation and depression
Outputs	Encoder values for rotation and a single H/L line for button depression
Functionality	The rotary dial works to scroll through menus, times and alphanumeric characters. The depression of the dial works to select.

Table 5.1.10 Menu Buttons Inputs, Outputs, and Functionality

Module	Menu Buttons
Inputs	Mechanical Depression
Outputs	Two High/Low lines
Functionality	The menu buttons are a back button for returning from a submenu and a utility button

Table 5.1.11 Dispense Buttons Inputs, Outputs, and Functionality

Module	Dispense Button
Inputs	Mechanical Depression
Outputs	One High/Low line
Functionality	Used for initiating a dispensing operation while in the dispensing sub-screen

Table 5.1.12 *Screen Inputs, Outputs, and Functionality*

Module	Screen
Inputs	On screen information
Outputs	Displays information for the user
Functionality	Displays information to the user based on the current position of the state machine.

User Interface Level 2:

The level 2 diagram of the user interface (Figure 5.1.4) consists of buttons, a dial, and a screen. The buttons and dial allow the user to control the machine and the screen. The screen displays information and settings for the user such as the dispensing configuration and time. The rotary dial and menu buttons allow the user to traverse and select settings from the screen. The MCU reads state changes from these inputs when the buttons are pressed, or the rotary dial changes position. The device controller block (Figure 5.1.2) responds by changing the screen accordingly.

The Screen is driven by an integrated LCD controller, the ILI9341V. The ILI9341V supports multiple connection modes. It supports 3 and 4 wire SPI interfaces as well as 8080 parallel interfaces with 8, 9, 16, and 18 bit buses. This project connects the display to the Microcontroller through SPI. This SPI connection contains the lines MISO, MOSI, SCLK, and CS. Even though the Screen has a D/CX line to tell the controller if the byte being sent is a command or data. The Screen also has a reset signal controlled by the microcontroller for resetting before initialization. In order to select this as the input mode, the input select pins IMO0, IMO1, IMO2, and IMO3 are set to low, high, high, and high respectively. This sets the display driver IC into 4-wire SPI mode II. The LCD backlight is driven at 12 volts while the logic is driven at 3.3V.

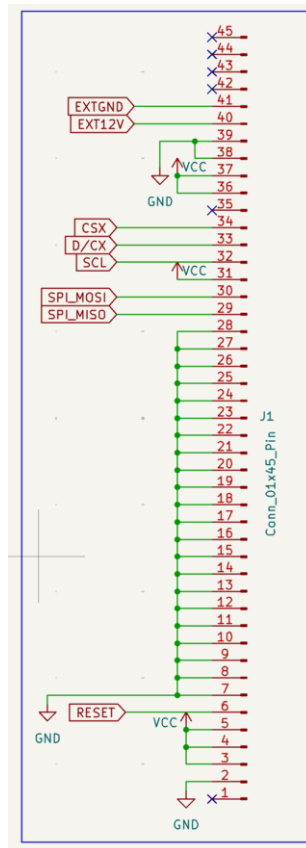


Figure 5.1.7 *Screen Connection Schematic*

The buttons are single pole single throw momentary push buttons. The switches chosen have a lot of switch bouncing, where the output of the switch quickly jumps between high and low logic levels many times at the moment it is pressed. The switches are connected to an RC circuit to convert this high frequency switching into a slow rise and fall between high and low logic levels. By connecting them to an RC circuit they are debounced. The signal however does not have sharp rising and falling edges. A Schmitt trigger inverter is a logic gate that requires the input signal to reach a threshold before it circuit changes the output signal. The debounced signal is fed into a Schmitt trigger buffer to give the microcontroller a single sharp rising edge to trigger off.

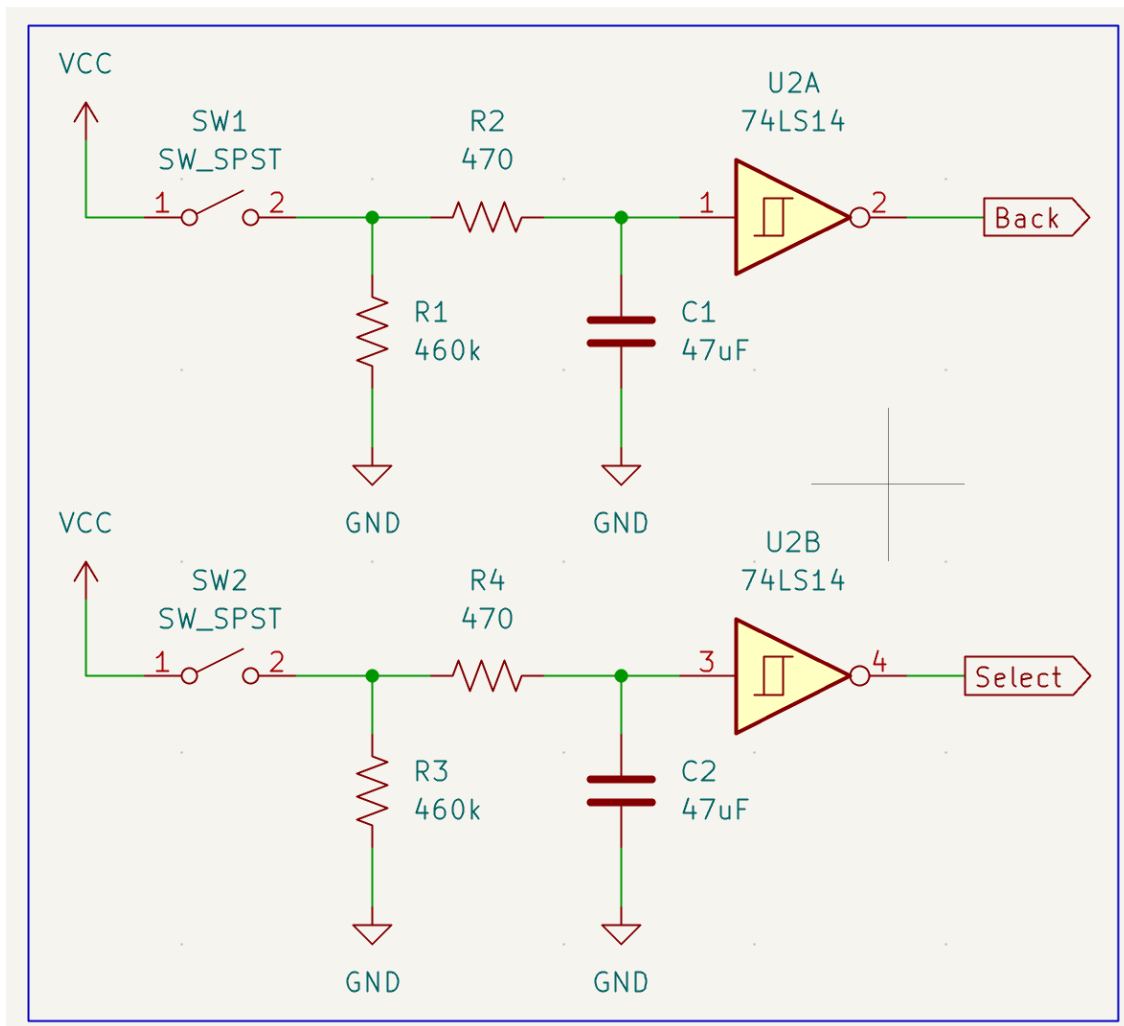


Figure 5.1.8 *Button Debounce Schematic*

The encoder has three pins A,B,C. Pins A and B are connected to pins on the Microcontroller and Pin C is connected to 3.3V. As the encoder spins pins A and B go high and low at different times. For clockwise rotation, pin B goes high before pin A. The opposite is true for counter-clockwise rotation. Through testing it was determined that the encoder did not have debouncing issues.

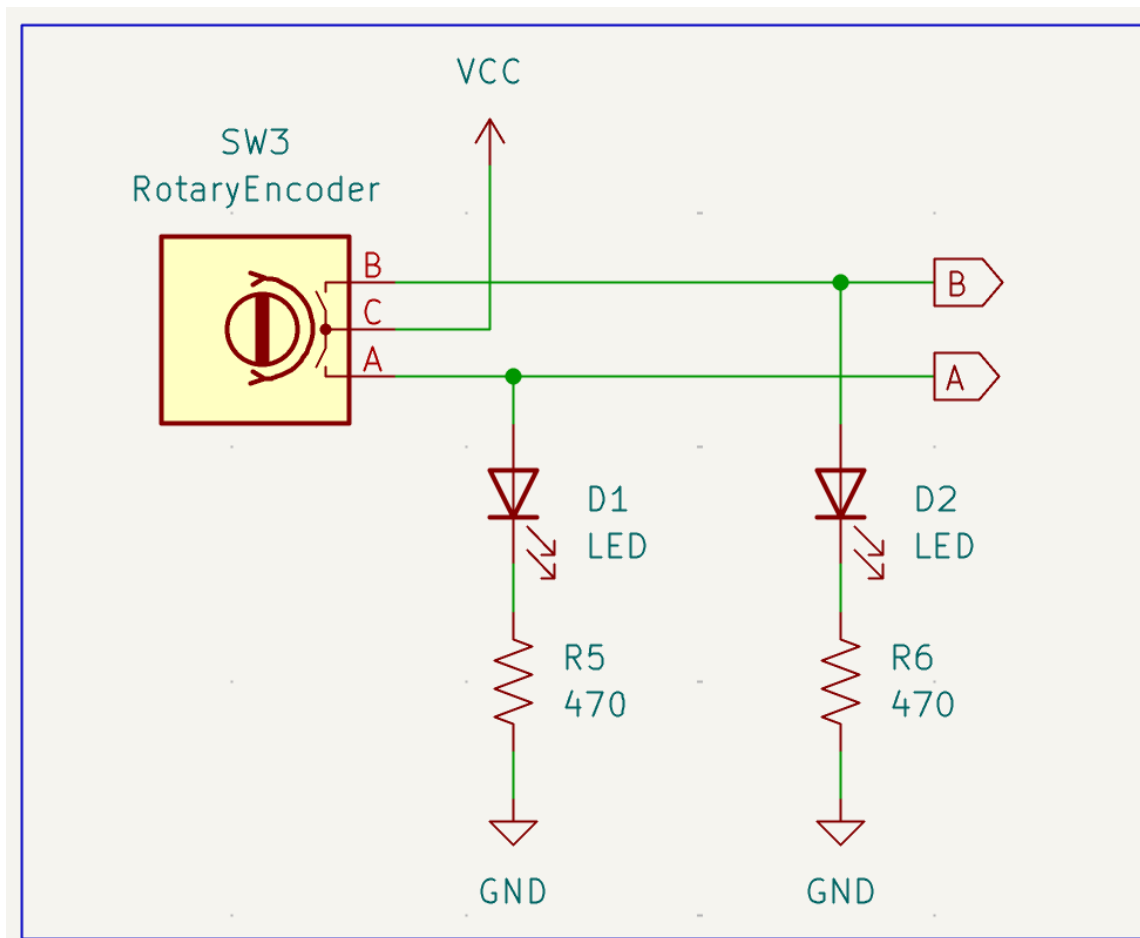


Figure 5.1.9 *Encoder Connection Schematic*

Authors: CB

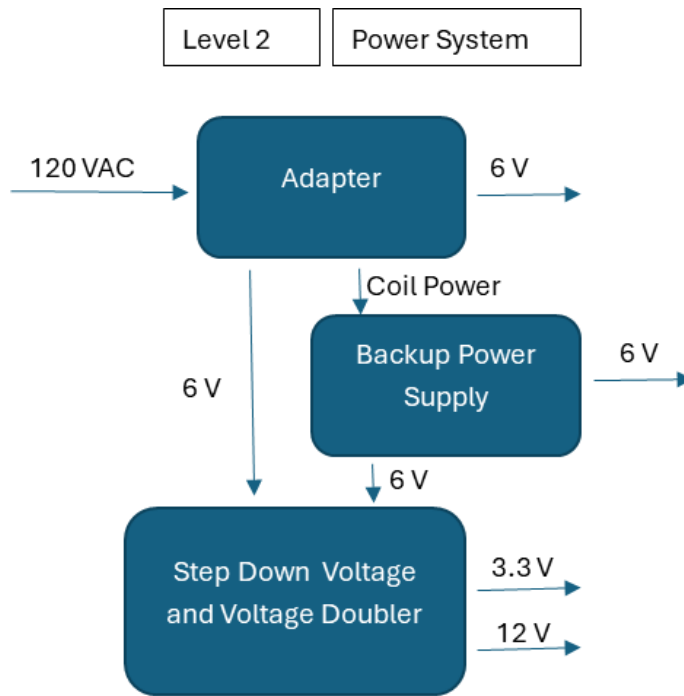


Figure 5.1.10 Level 2 Functional Diagram of Power System

Table 5.1.13 Adapter Inputs, Outputs, and Functionality

Module	Adapter
Inputs	120 V AC
Outputs	6 V DC
Functionality	Acts as the main power source for the device.

Table 5.1.14 Backup Power Supply Inputs, Outputs, and Functionality

Module	Backup Power Supply
Inputs	Power from Adapter
Outputs	6 V
Functionality	If the device loses power from the adapter, the Backup Power Supply will supply power to the device via a battery.

Table 5.1.15 Step Down Voltage Inputs, Outputs, and Functionality

Module	Step Down Voltage and Voltage Doubler
--------	---------------------------------------

Inputs	6 V from Input Power or Backup Power Supply
Outputs	3.3 V and 12 V
Functionality	Converts 6 V to 3.3 V and then doubles 6 V to 12 V

Power System Level 2:

The power system level 2 functional diagram (Figure 5.1.5) describes the process of converting the input power from the wall plug to providing power to the rest of the subsystems. The adapter (Table 5.1.13) inputs 120 V AC from the wall and outputs 6 V DC. The adapter outputs 6 V DC to the backup power supply, the Step Down Voltage module and the rest of the system. The next module is the Backup Power Supply (Table 5.1.14). This system detects loss of power from the adapter and switches the circuit to be battery powered. It outputs 6 V to the Step Down Voltage and Voltage Doubler and the rest of the system. Lastly, the Step Down Voltage and Voltage Doubler module converts the 6 V input to 3.3 V to power the device's lower-voltage components and doubles the 6 V to 12 V to power the display's backlight.

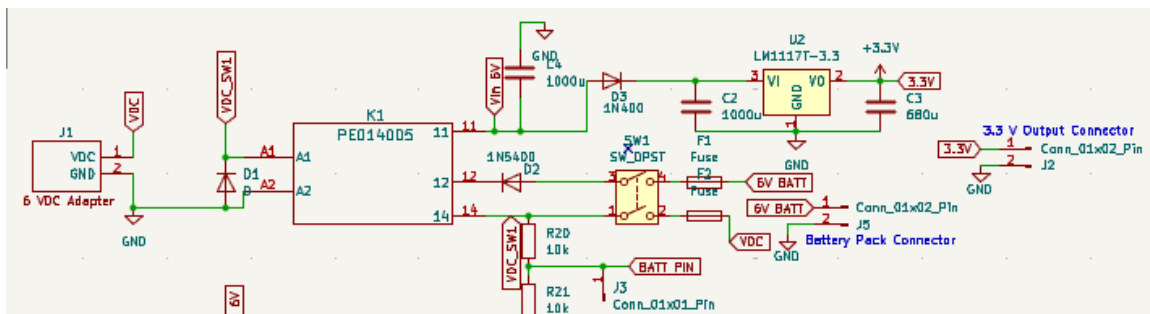


Figure 5.1.11 Backup Battery Circuit Schematic

The hardware implementation for the Backup Power Supply module is shown as a schematic in Figure 5.1.6. The design uses a double-throw relay to switch the circuit from being powered by the power adapter to being powered by the battery. The coil of the relay is powered by the adapter so when the system loses power from the adapter, the relay switches the circuit to the 6 V battery system. This circuit also has a DPST master switch that will disconnect both the battery pack and the 6 V adapter from the system

when the device needs to be completely powered off. The circuit uses a linear regulator (LM1117T-3.3) to step down the 6 V to 3.3 V that will be going to the MCU board.

Author: RL

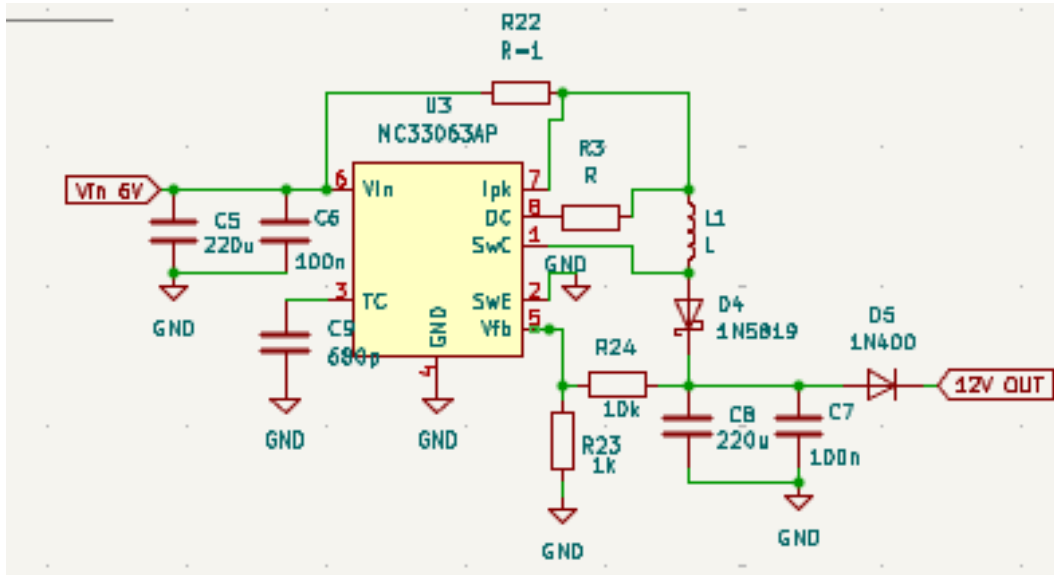


Figure 5.1.12 12V Boost Converter Circuit Schematic

To implement the 12 V input needed by the display backlight, a boost converter is used as shown in the schematic above in Figure 5.1.7. This system uses the NC33063AP chip to input 6V and output around 12.6 V. This voltage is slightly higher than the required voltage but still below the maximum back light voltage. Originally, the circuit design used a voltage doubler with only passive components. When testing the doubler system with the rest of the power system, the voltage sags, which causes the screen to flicker when servos are pulling from the power supply. The boost converter fixed the voltage doubler problem and supports the screen demand while the servos are moving through the dispense cycle.

Author: RL

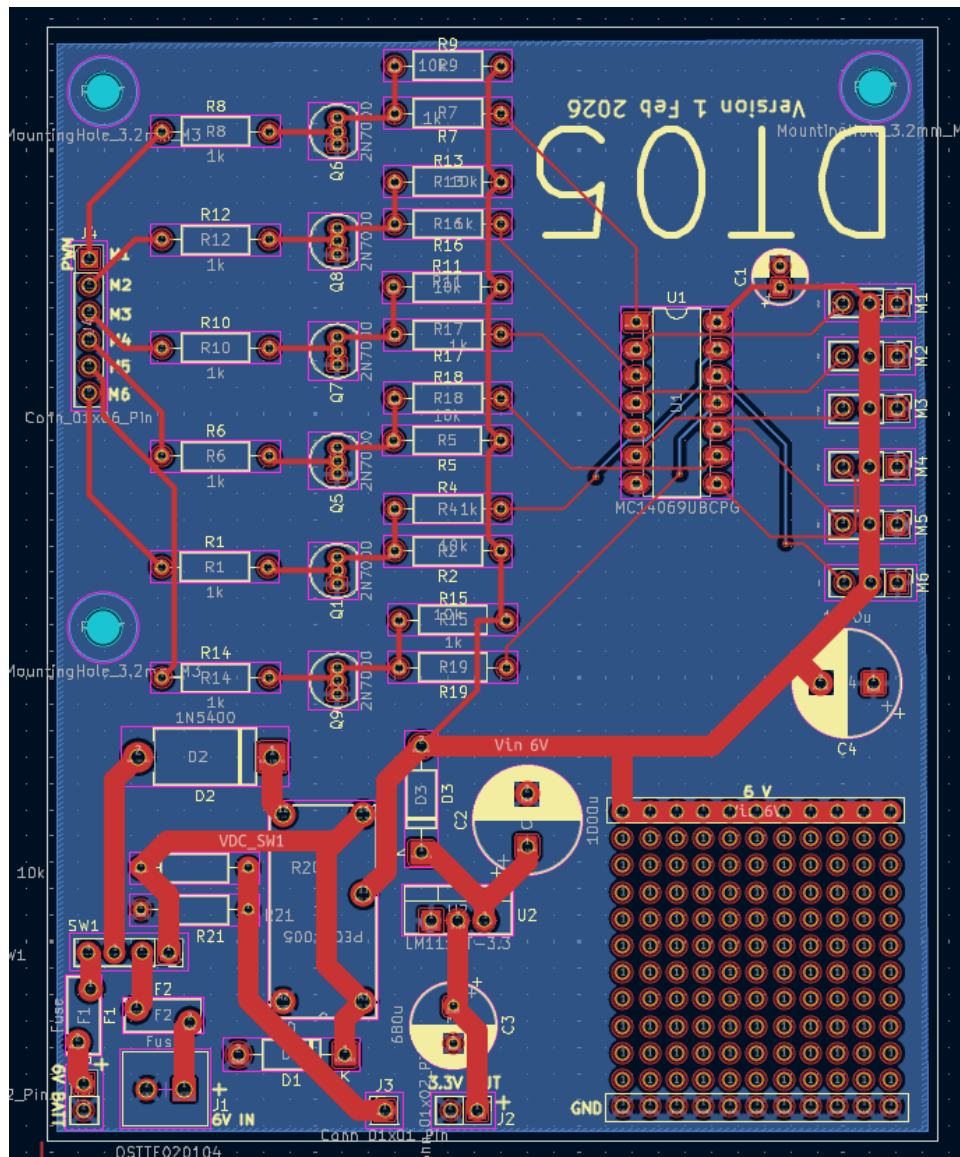


Figure 5.1.13 *Power Board PCB Layout*

The PCB layout includes the entire power system and level shifters for servo control. The relay, 3.3 V regulator and boost converter perf board are at the bottom of the board, and the level shifters for the servo control are at the top. When designing this layout, the 12 V circuitry was still not fully developed so the perf board section that has a 6V and GND line allows space for the boost converter components. These 6 V connections also allowed connections for the LEDs and phototransistors used for dispensing verification. The board has connectors for the 6 V battery pack, master switch, 6 V adapter input, 3.3 V output, 12 V output, six inputs for the MCU PWM signals, and six servo connections. The PCB has two copper layers; the top layer (Red) is primarily used for all the connection routing, and the bottom layer (Blue) is a ground plane with a

few exceptions for routing the inverter chip. All components are through-hole for simplicity and to use the parts used for prototyping.

Author: RL

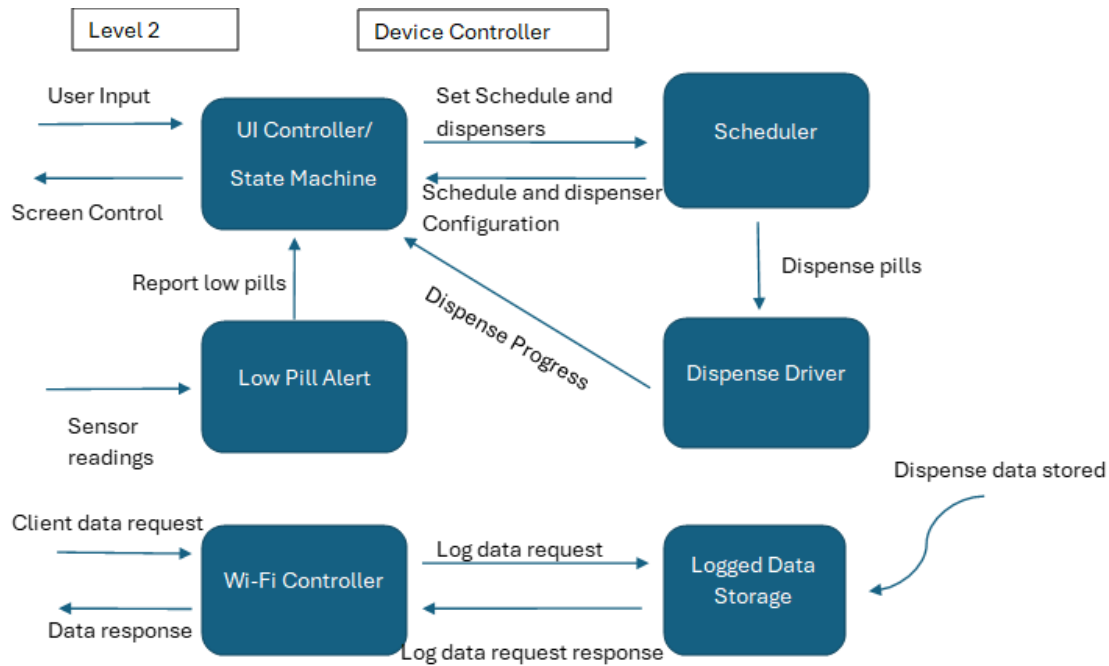


Figure 5.1.14 Level 2 Functional Diagram of Device Controller

Table 5.1.16 Scheduler Inputs, Outputs, and Functionality

Module	Scheduler
Inputs	Takes in user input in UI
Outputs	Dispense trigger
Functionality	Keeps track of time and the dosage schedule to send out signals when pills need to be dispensed

Table 5.1.17 Wi-Fi Controller Inputs, Outputs, and Functionality

Module	Wi-Fi Controller
Inputs	Client data requests, Log data request response
Outputs	Data response, log data request
Functionality	Receives internet requests for log data. If valid requests, retrieves data from non-volatile memory and responds

Table 5.1.18 Logged data Storage Inputs, Outputs, and Functionality

Module	Logged data storage
--------	---------------------

Inputs	Log data requests
Outputs	Log data request response
Functionality	Stores data logged from machine usage

Table 5.1.19 *UI Controller / State Machine Inputs, Outputs, and Functionality*

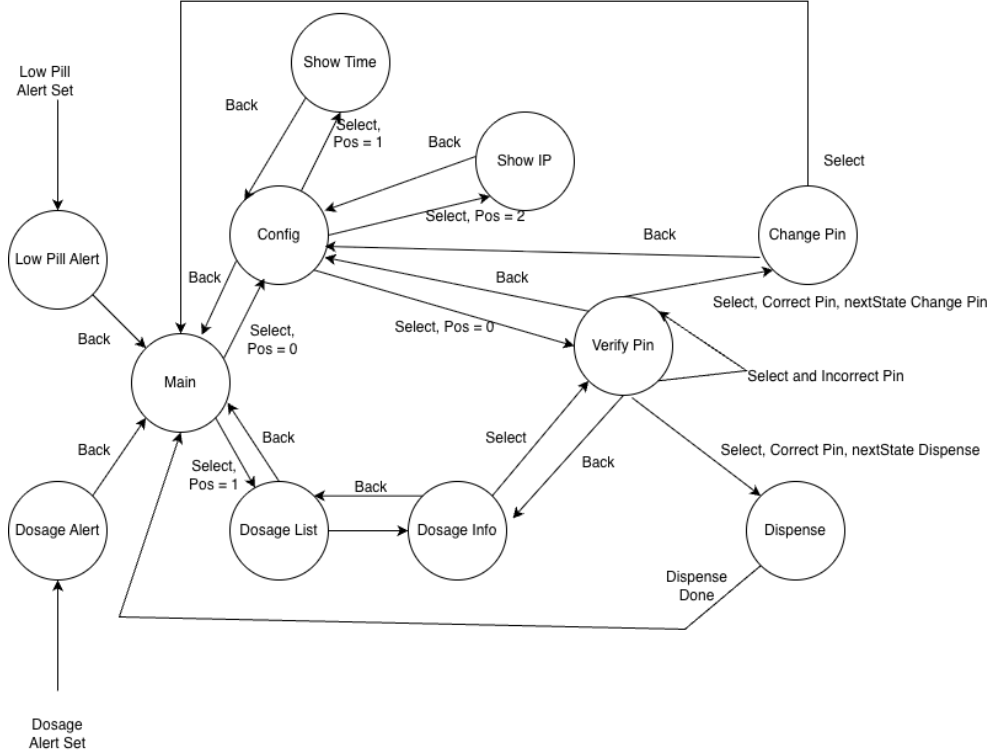
Module	UI Controller / State Machine
Inputs	User input from User Interface Dispense Progress Scheduler Configuration
Outputs	Scheduler configuration update information Dispenser configuration update information Current config data to UI interface
Functionality	Talks back and forth between the UI and the various parts of the device controller. Below is a state machine diagram that shows each state and their various interconnections.  <pre> stateDiagram-v2 [*] --> Main Main --> Config : Back Main --> DosageList : Back Main --> LowPillAlert : Low Pill Alert Set Main --> DosageAlert : Dosage Alert Set Config --> ShowTime : Back Config --> ShowIP : Back Config --> ChangePin : Back Config --> VerifyPin : Back Config --> Main : Select, Pos = 0 ShowTime --> Config : Select, Pos = 1 ShowIP --> Config : Select, Pos = 2 ChangePin --> Config : Back ChangePin --> VerifyPin : Select, Correct Pin, nextState Change Pin VerifyPin --> Config : Select, Pos = 0 VerifyPin --> Main : Select and Incorrect Pin VerifyPin --> Dispense : Select, Correct Pin, nextState Dispense DosageList --> Main : Back DosageList --> DosageInfo : Back DosageInfo --> VerifyPin : Select Dispense --> Main : Dispense Done </pre>

Table 5.1.20 *Dispense Drive Inputs, Outputs, and Functionality*

Module	Dispense Driver
Inputs	Pills to dispense Dispense Sensor
Outputs	Dispense Progress Servo control signals
Functionality	Takes in a number of pills to dispense from each of the containers and handles reading sensors and controlling dispenser hardware

Table 5.1.21 *Low Pill Alert Inputs, Outputs, and Functionality*

Module	Low Pill Alert
Inputs	Container sensors
Outputs	Low pill alerts
Functionality	Takes in the container sensors to determine if the containers are low on pills and sends signals to alert the user accordingly.

Device Controller Level 2:

The device controller block diagram (Figure 5.1.7) handles all operations related to processing UI inputs and writing UI screen outputs, determining when to dispense pills, coordinating pill dispenses, determining when pill containers are low, and receiving/responding to data requests over Wi-Fi. The UI controller receives inputs from the physical inputs and determines the appropriate screen response. The physical button inputs and screen are described in Figure 5.1.4. The state machine that reflects UI control can be seen in table 5.1.19. The scheduler block is responsible for keeping track of when the pills are to be dispensed and in what quantity. When the scheduler determines it is the appropriate time for a dispense to occur, it notifies the dispense driver that it need to dispense the configuration of pills specified by the user. The dispense driver block is responsible for controlling the servos and sensors that collect pills from their containers and dispense them. The servos are run until a pill is sensed in the dispensing chute. This operation is repeated until all the pills to dispense are collected in the chute. When all pills are collected, the final gate is opened, dispensing them for the user to collect. The low pill block is responsible for reading sensors in pill containers and determining if the

pills have gone below a certain level. The logged data storage block is responsible for storing information related to pill dosage, times, quantities, and missed dosages. This block will receive requests to store and retrieve this data. The Wi-Fi block is responsible for receiving and handling all data requests. The block will receive a request, retrieve the appropriate data from the logged data block, and send it to the Wi-Fi client that requested it.

The hardware implementation for this block will consist of an STM32 microcontroller which will read buttons and sensors from the device and activate servos. It will interface with a Wi-Fi chip, Flash chip, and an LCD screen through serial communication.

Authors: RO, CB

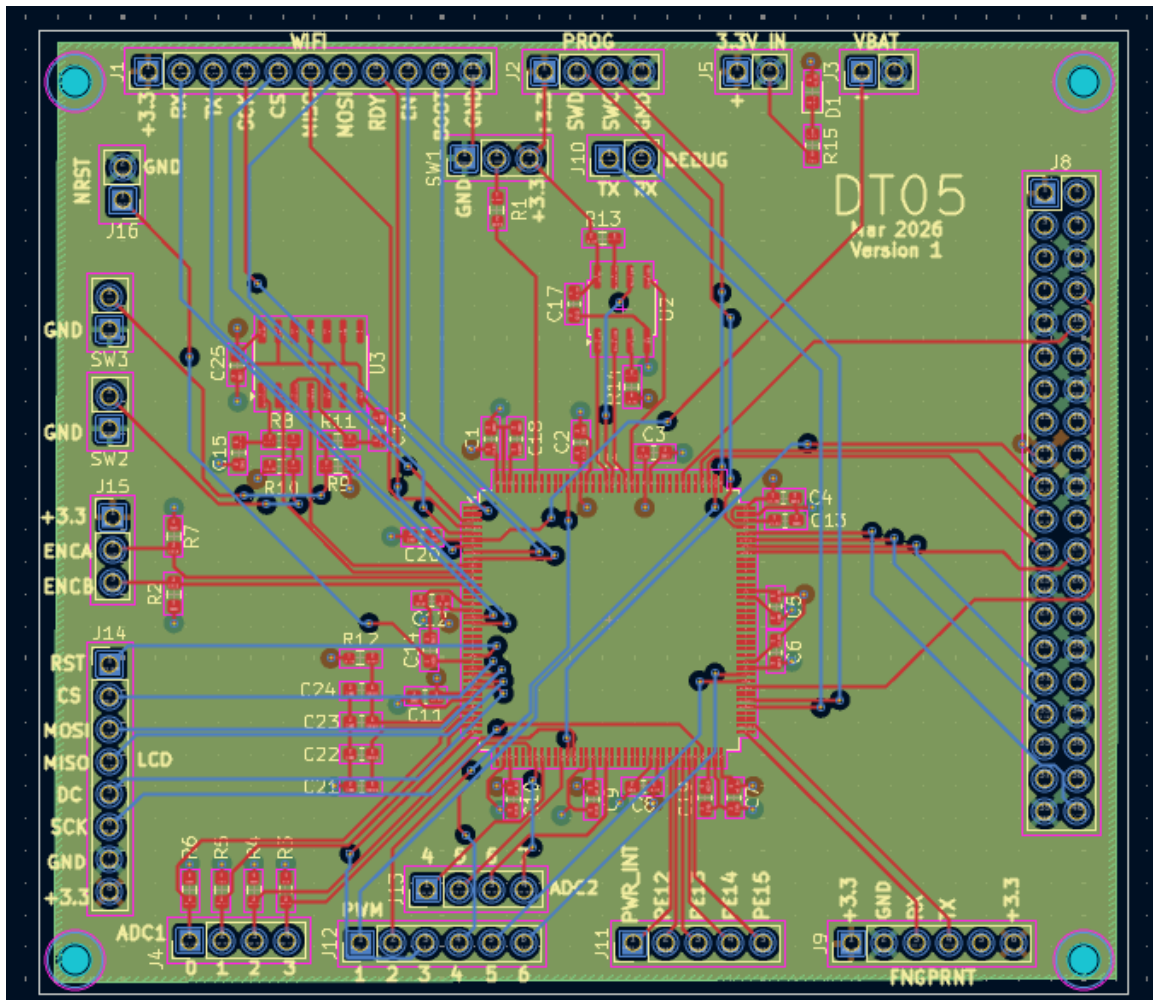


Figure 5.1.15 Microcontroller PCB Layout

The MCU PCB is a four-layer PCB with signals on the top and bottom layers and 3.3 V power and ground plane in between. Figure 5.1.15 shows the PCB layer in KiCAD. The board features the MCU chip in the center, the denouncing circuitry for the UI controls, flash chip and multiple through-hole connections to control all sub systems. The board is made to directly connect to the 3.3 V connection on the power board. The board connects directly to the WiFi module board through the J8 header. This direct connection eliminates signal integrity issues found during prototyping and development. This board is designed with extra header connections for unexpected challenges that arise during testing. J9, J11, J13 and J1 are all backup headers that are not directly included in the design. Some of these backup headers were used in the final product.

Authors: RL

5.2. Software Design:

The “device controller” block in the functional breakdown diagram is responsible for software operations that will: Control the UI, maintain a pill dispensing schedule, notify the user when a pill container is low, coordinate dispenses, receive and respond to requests for data, and store log data locally. These are level 2 block functions within the device controller.

UI Controller/ State Machine Level 2

The UI controller is a state machine that allows the user to input data into the machine and extract information about its current setup. The state diagram in table 5.1.19 shows 7 different states that the UI controller can be in, each with its own menu that displays data to the user. The switching between these menus is achieved using two push buttons and rotary dial. The two push buttons are back and select. The back button always moves backwards in the state machine towards the default menu. The select button will select whatever is highlighted on the screen. the radial dial’s function will be determined based on the different menu. In the default, config, schedule and dispense menus, the radial dial will scroll between available options. The select button in these menus will move the user into the submenu that is highlighted. The radial dial will serve two functions in both dispenser and dose config menus. Upon loading into the menu, the radial dial will scroll between different values that can be changed by the user and a save button. Upon selection of one of these values, the radial dial will be able to be turned to alter the value up or down. Pressing select again will change the dial back to scrolling between the different values in the menu. When the save button is selected in this menu, the values on the screen are sent to the scheduler to be saved for use. In the dispense menu, a dosage can be dispensed at any time.

SPI communication is used to interface with the screen. Many SPI drivers for the ILI9341 LCD Driver exist. By repurposing one made for the STM32F4 series of microcontrollers, a core set of commands can be sent to the device driver that are translated into SPI commands (Eziya). This code is universal between the STM32F4 and STM32H5 microcontrollers due to the similarities in the Hardware Abstraction Layer. The code for SPI communication however does not directly translate between the STM32F4 and the STM32H5. To accomplish SPI communication, the functions responsible for SPI transmission had to be rewritten for the STM32H5. Another goal for

the screen was to use the direct memory access (DMA) controller. The DMA controller allows for the microcontroller to pass in a buffer to be transmitted rather than individual bytes. The code used to accomplish the SPI transmissions is below. It takes in either a single byte or a buffer to send over SPI.

```
static void ILI9341_SPI_Tx(uint8_t data)
{
    // Check GPDMA busy Flag
    while((GPDMA1_Channel2->CSR & 0x100));
    HAL_SPI_Transmit_DMA(HSPI_INSTANCE, &data, 1);
    //HAL_SPI_Transmit(HSPI_INSTANCE, &data, 1, 10);
}

static void ILI9341_SPI_TxBuffer(uint8_t *buffer, uint16_t len)
{
    // Check GPDMA busy Flag
    while((GPDMA1_Channel2->CSR & 0x100));
    HAL_SPI_Transmit_DMA(HSPI_INSTANCE, buffer, len);
    //HAL_SPI_Transmit(HSPI_INSTANCE, buffer, len, 10);
}
```

Figure 5.2.1 *SPI transmission code snippet*

The inputs are handled through interrupts. The interrupts call a callback function that sets a variable to indicate that the state of the input has changed. For the two pushbuttons, this is a binary state that is set to 1 when the button is pressed and is cleared in the main loop when action is taken based on the state. For the encoder, the state of the encoder can be determined with only one interrupt. The interrupt is set on the A pin of the encoder. It reads the B pin to determine which direction the encoder is turning. If pin B is high, then the encoder is turning clockwise. If pin B is low then the encoder is turning counterclockwise. These states are defined in an enumeration.

```
enum encMovement{
    NONE, // no movement
    CW, // Clockwise Movement
    CCW // Counter Clockwise Movement
};
```

Figure 5.2.2 *Encoder movement enumeration*

```

volatile unsigned short selectPressed;
volatile unsigned short backPressed;
volatile enum encMovement encoderState;

void selectButtonCallback(){
    selectPressed = 1;
}

void backButtonCallback(){
    backPressed = 1;
}

// called on rising A transition
// Assume A is high
void ENCACallback(){
    // check to see if B is already high
    if(readB)
        encoderState = CW;
    else
        encoderState = CCW;
}

```

Figure 5.2.3 *Button and Encoder code snippet*

The UI state machine updates based on user input. It takes in UI events and the current UI state and updates the UI state based on these, then redraw the screen if necessary. Pseudocode for this functionality looks like.

```

void UIEvent(UIState& state, inputAction input){
    uint8_t redraw = 0;
    switch(input){
        case SELECT:
            switch(state):
                ...
            break;
        case BACK:
            switch(state):
                ...
            break;
        ...
    }

    if(redraw){
        redrawScreen(state);
    }
}

```

Figure 5.2.4 *UI state machine pseudocode*

A pin code is used for verification. It is a secret code of 4 numbers to verify the user and add security to the dispenser. The pin code verification is set up as a special state in the user interface state machine. There are two functions that come along with the pin code system. These are pin code verification and pin code setting. To set a pin code the user must first validate themselves through pin code verification and then they can set the pin code. To select a pin code, the user move the encoder clockwise and counter-clockwise to increment and decrement the currently selected place. The select and back buttons are used to move through the different places of the pin code. The pin code only allows selection of the characters from '0' to '9'. The pseudo code for implementing this functionality is as follows.

```
void handleEncoder(bool dir){
    if dir {
        increment selected character;
        if selected character > '9'{
            selected character = 0;
        }
    }
    else {
        decrement selected character;
        if selected character < '0'{
            selected character = 0;
        }
    }
}

void handleSelect(){
    increment pin index
    if pin index > 3{
        go to next state;
    }
}

void handleBack(){
    decrement pin index;
    if pin index < 0 {
        go to back state;
    }
}
```

Figure 5.2.5 *pin code pseudocode*

Scheduler Level 2

The scheduler saves dosage information passed to it from the user interface and combined with the real time clock can correctly alert the user to a dose. The scheduler checks every minute to see if it is time to take a dosage. When it is time, dispenser alerts the user and changes the display to the appropriate dosage information.

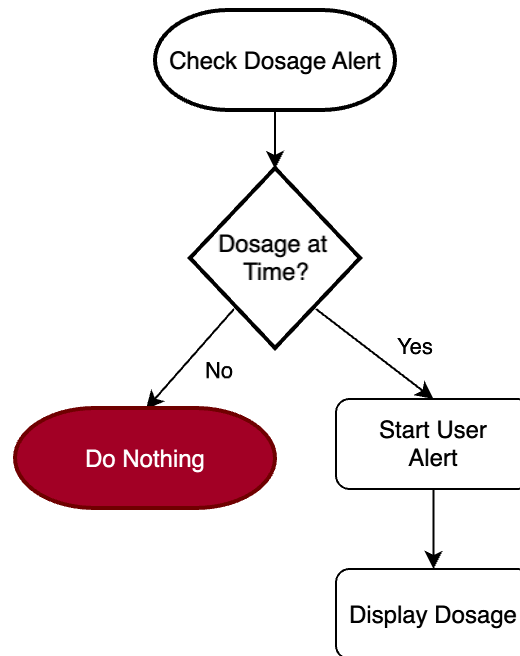


Figure 5.2.6 *Flow chart for scheduler operation*

Authors: CB

Low Pill Alert Level 2

The low pill alert will use pill counts to determine when pill counts go under a certain threshold. Pill counts are configured with a pill channel and are decremented every time a pill is dispensed from a given channel. Periodically, this variable is checked and if a flag has not yet been raised it will give out an alert to the user.

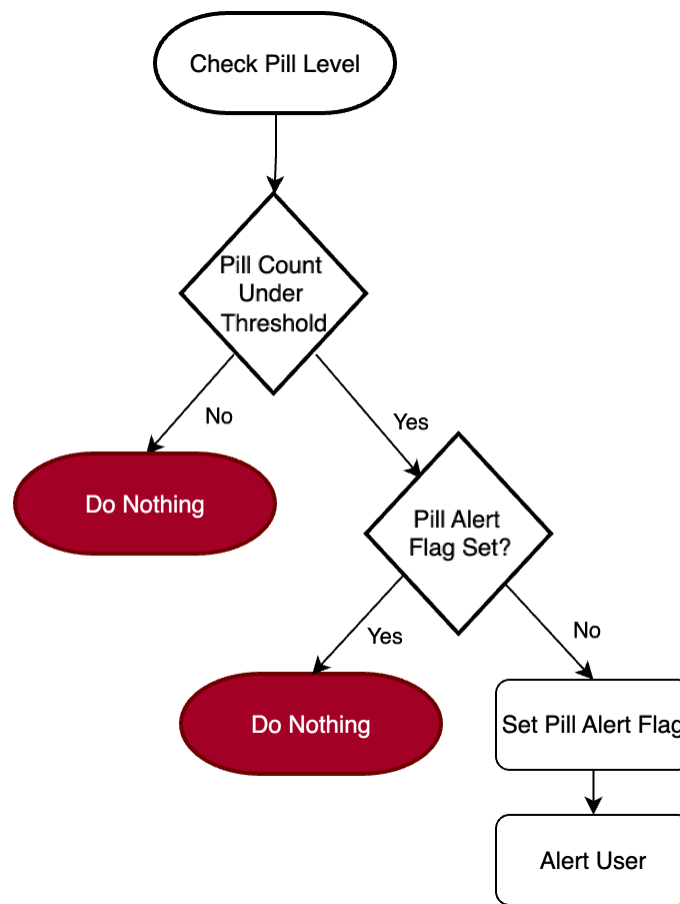


Figure 5.2.7 *Flow chart for checking for low pill*

Authors: CB

Dispense driver Level 2

The dispense driver is responsible for controlling servos and reading from sensors to coordinate a successful dispense of pills. The dispense driver will be notified when it is time to dispense along with which pills to dispense. A successful pill dispense will look like running the servos to grab a pill and release it into the dispensing chute. A pill "pick" has successfully occurred by software definition when the sensor in the dispensing chute detects a pill has passed through. This sequence will repeat until the desired amount and variety of pills has been collected in the dispensing chute. Once all pills have been collected, the gate will open for the user to access. The gate will be controlled by another servo. Figure 5.2.3 demonstrates the flow of a pill-dispense operation.

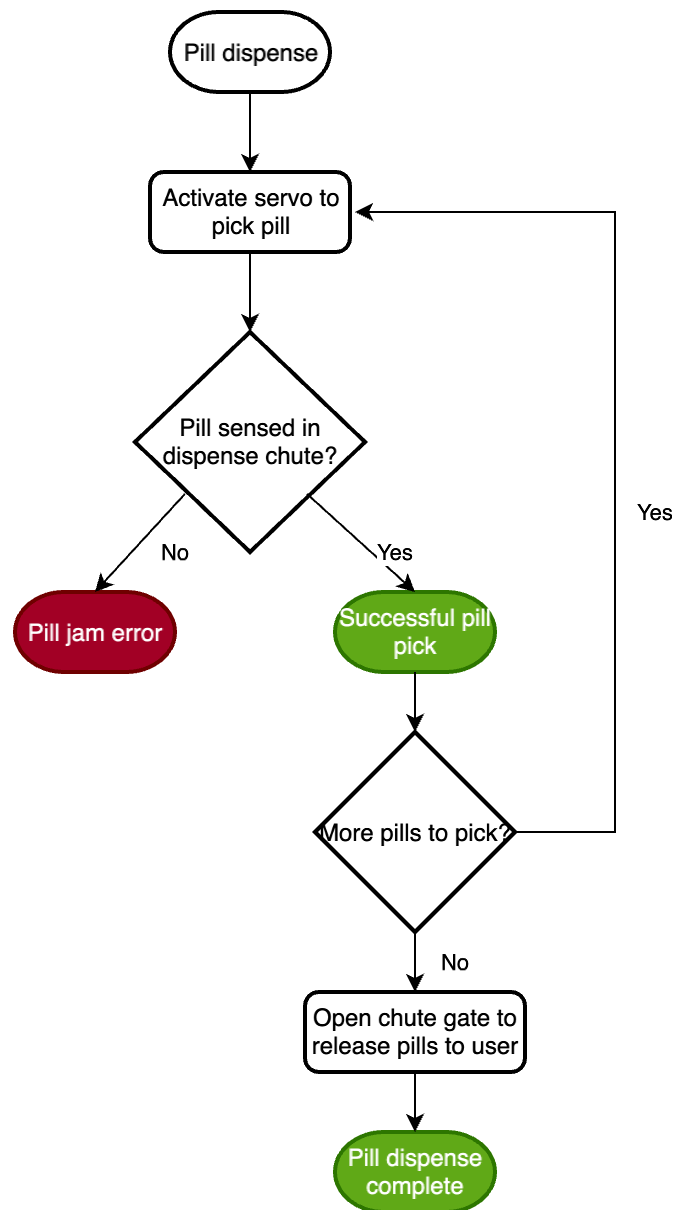


Figure 5.2.8 *Flow chart for dispensing operation*

The software for the dispense driver involves controlling servos and reading feedback from a phototransistor. The dispense operation will consist of moving servos to 'pick' a pill and using a phototransistor to determine when a pill has fallen through the dispensing chute.

The servo control code moves the servos to any position provided. The pill dispenser requires operating various types of servos to handle pill dispense operations. The sequence of dispensing pills involves running 270-degree servos to pick one pill at a time. Once all the pills are picked, a 180-degree servo is used to dispense the final collection of pills to the user.

Servos take 3 inputs: VCC, GND, and Control. The control line is a PWM signal that determines the speed or degree (continuous servos are controlled with speed, 180- and 270-degree servos are controlled in degrees). The servo PWM for this project operates at a 20ms period with pulses ranging from .5ms to 2.5 milliseconds. The table below describes how various pulses impact the operation of servos:

	.5 mS	1.5 mS	2.5 mS
Continuous Servo	reverse fastest	no movement	forward fastest
180 Servo	0 degrees	90 degrees	180 degrees
270 Servo	0 degrees	135 degrees	270 degrees

Table 5.2.1 *Servo control breakdown*

In a continuous servo, the smallest pulse corresponds to the reverse direction at the fastest speed. The middle pulse, 1.5ms, corresponds to no movement. The longest pulse, 2.5, corresponds to the fastest speed in the forward direction. The 180- and 270-degree servo responds to varying pulses by moving to different degrees. The MCU generates PWM signals using a timer peripheral that is configured to 'PWM mode' with frequency and pulse width.

The functions developed to drive the servos are:

```
int runContinuousServo(TIM_HandleTypeDef timer, uint32_t tim_channel, int speed);
int run180Servo(TIM_HandleTypeDef timer, uint32_t tim_channel, uint32_t degree);
int run270Servo(TIM_HandleTypeDef timer, uint32_t tim_channel, uint32_t degree);
int stopServo(TIM_HandleTypeDef timer, uint32_t tim_channel);
```

Figure 5.2.9 *Function declarations to control servos*

The run functions compute the appropriate pulse width required for the speed or degree provided and apply it to the specified timer peripheral (specified by timer and tim_channel). The runContinuousServo function takes a parameter speed, which ranges on [-100, 100]. -100 is the fastest reverse, 0 is neutral, 100 is fastest forward. The equation that this function uses to determine the pulse width is:

$$pulseSize = 1.5ms + percentSpeed * 1ms$$

If the percent speed is -100, the computed pulse size is .5ms. If the percent speed is 100, the calculated pulse size is 2.5ms. The run180Servo and run270Servo functions

work similarly but take parameters from 0 to 180/270. The equation for these functions are:

$$pulseSize = .5ms + degree/180 * 2ms$$

$$pulseSize = .5ms + degree/270 * 2ms$$

The stopServo function disables the PWM signal on the pin specified. It can be applied to all servos regardless of their type.

The 'feedback' portion of the dispense driver is responsible for determining when a pill has been picked. When a pill has been successfully 'picked', a pill will pass between a phototransistor and led. When the pill passes through, the phototransistor will report a drop in brightness. The software will 'listen' for these events to notify when a pill has been passed through the chute.

The software implements this functionality with external interrupt pins configured to trigger on a rising edge. The interrupt triggers when a pill passes phototransistor, and the flag will be set to indicate that a pill has been detected. Pseudocode for this functionality will look like:

```
void Interrupt_Callback(uint16_t GPIO_Pin) {  
    if(GPIO_Pin == PhotoTransistorPin)  
    {  
        pillDetected = true;  
    }  
}
```

Figure 5.2.10 Pill picked flag pseudo code function definition

The servo drivers and phototransistor feedback will be combined to coordinate dispenses. Pseudocode to illustrate this looks like:

```
bool pickPill() {  
    // Move servo to 'home', pick pill  
    run270Servo(servo1, 0);  
  
    // Move servo to first dispensing slot  
    run270Servo(servo1, 135);  
    sleep(500);  
  
    if (pillDetected)  
        return true
```

```
return false;  
}
```

Figure 5.2.11 *Pick pill pseudo code function definition*

This code snippet illustrates a sequence of trying to pick a single pill. The servo moves the pill container to a slot where a pill can be dropped. If a pill is successfully dropped through, the function returns true. If no pills were detected falling through the chute, the function returns false.

Authors: RO

Wi-Fi Controller Level 2

The Wi-Fi controller is responsible for receiving and responding to requests for log data. Other devices will send a data request to the pill dispenser, and a response will be built with the requested data.

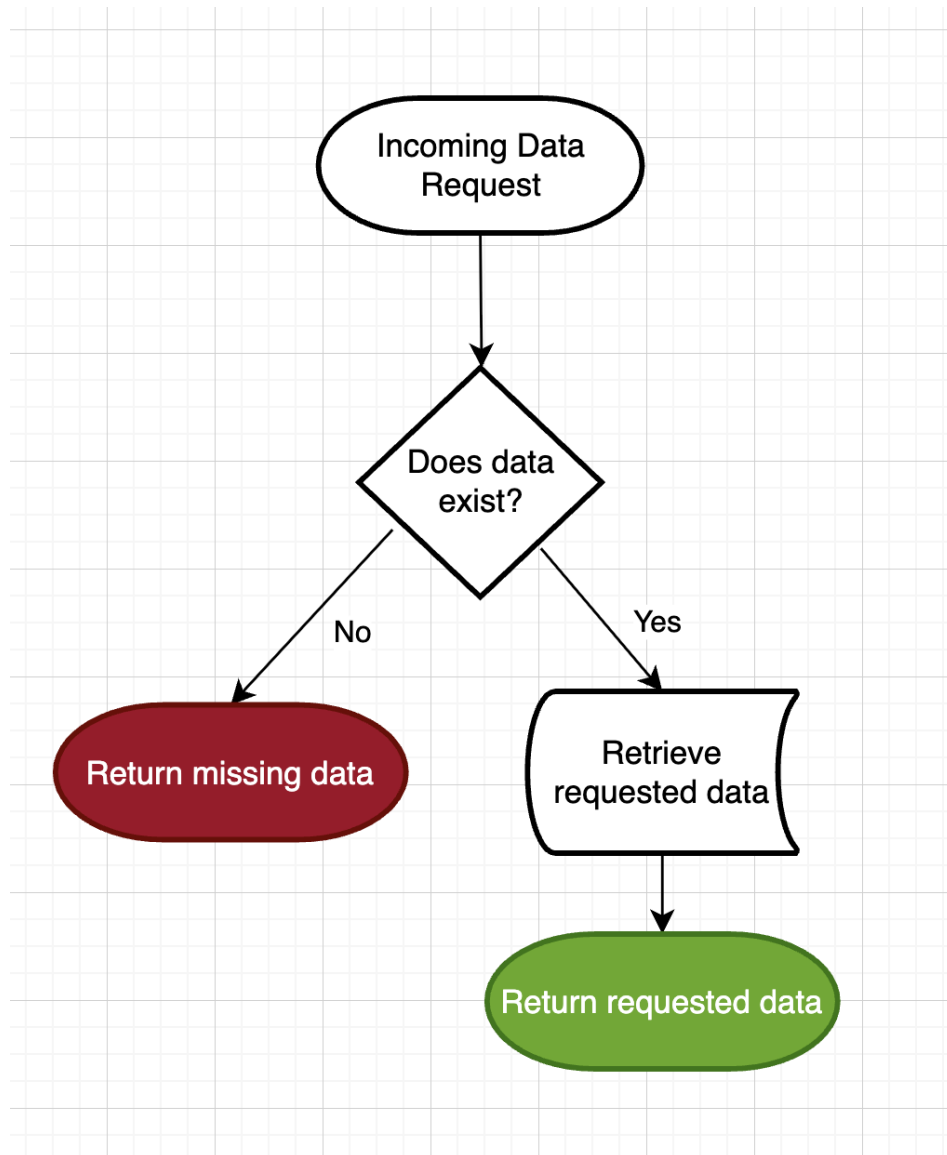


Figure 5.2.12 Flow chart for Wi-Fi module operation

The ST67W611M1 Wi-Fi chip will be used for internet connectivity. The MCU will send commands to the device via serial peripheral interface. The MCU receives HTTP messages, parses them to determine the type of request, performs the action request, and builds an appropriate HTTP response. High level pseudo code to demonstrate this idea looks like:

```
void WiFi_Message_Received_Callback(char* payload) {  
    int response_type = parsePayload(payload);  
  
    if (response_type == GET_LOG)  
        respond_with_log(payload);  
}
```

```

else if (response_type == GET_CONFIG)
    respond_with_config(payload);
else if (response_type == ...)
    respond_with_...(payload);
...
}

```

Figure 5.2.13 *Wi-Fi function definition*

Authors: RO

Logged Data Storage Level 2

The pill dispenser will save information, such as when pills were dispensed, what kind, missed dosages, etc. This data will be organized in compact fixed amounts of bytes to maximize space. This data will be stored in persistent memory on the pill dispenser. Modules such as the dispense driver and the scheduler are responsible for reporting data to this block. The Wi-Fi block module is the primary device that will request information from this block because it deals with external requests for data.

The hardware responsible for storing logged data is a flash chip. This chip is a form of nonvolatile storage that is operated via serial peripheral interface. The Flash chip will receive requests to store log data, and requests to retrieve log data. While not implemented, the functions for driving this chip will look like:

```

bool write_flash(uint32_t address, char[] log);
char[] read_flash(uint32_t address, uint32_t size);

```

Figure 5.2.14 *NOR Flash chip read/write pseudo-code function definitions*

The write function takes an array of characters and an address to specify where in the flash chip to write the data. The read function will return the data from the address specified. The implementation will involve communicating with the flash chip via SPI. The project plans to use a lightweight embedded filesystem, such as LittleFS, to create safer data storage that's resilient to power-loss.

Authors: RO

Microcontroller Connectivity Diagram

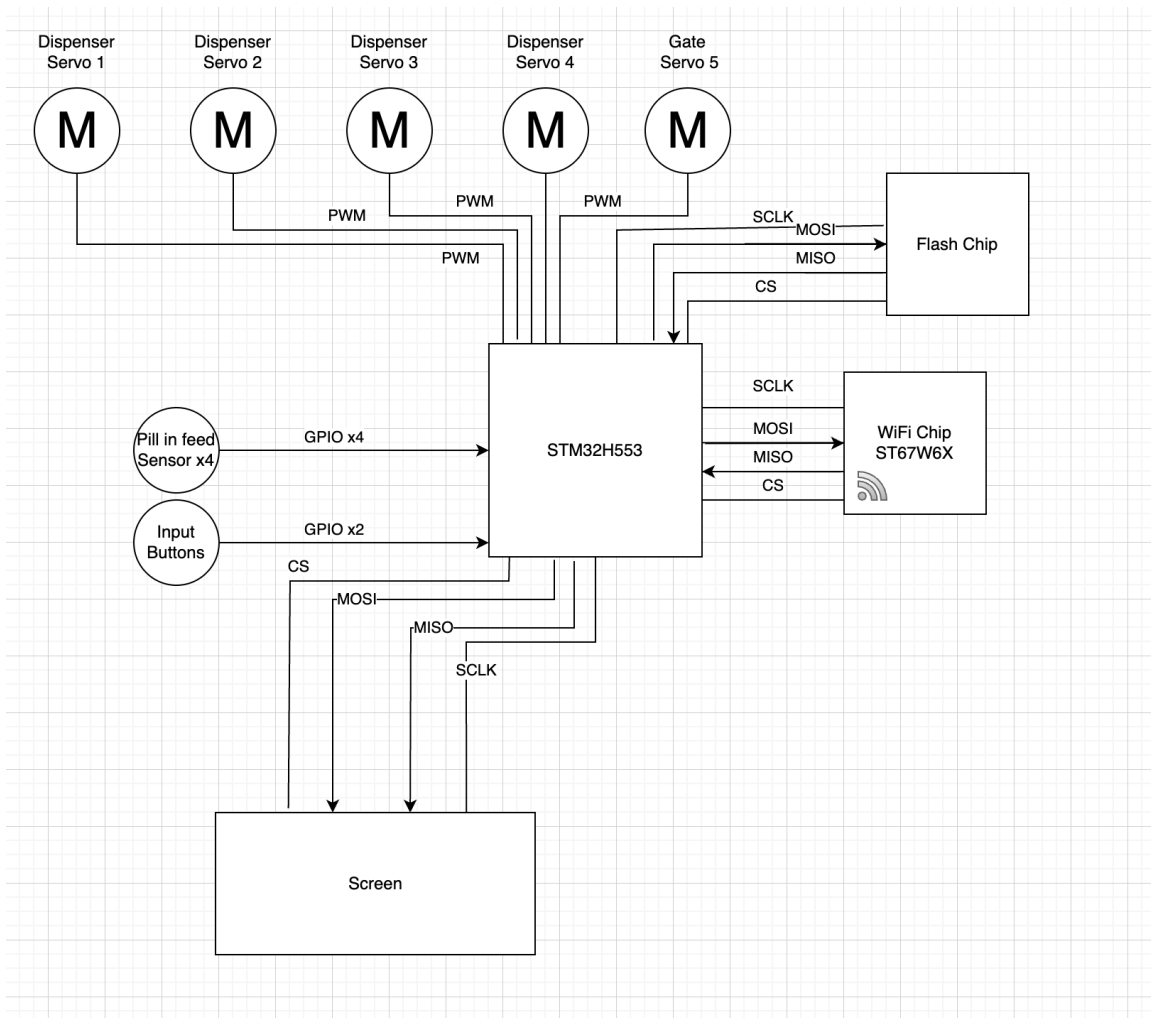


Figure 5.2.15 *Microcontroller Connectivity Diagram*

The diagram above shows all the project hardware connected to the MCU from a software perspective, and the type of communication used. The MCU communicates with hardware through pulse width modulation, general purpose input/output, or serial communication protocols (UART, SPI).

Subsystem Testing Procedures

The project was split into smaller subsystems to divide work and prove that each part of the project could work individually. The subsystems were power supply, pill dispensing, Wi-Fi connectivity/logged data storage, and user interface.

Power Supply

The power supply system was continuously tested throughout the design process. The power system is designed based on the entire demand of all of the subsystems, so it

depends on design requirements outside of the power subsystem. As more subsystems were integrated, more power tests took place.

The first components tested on the power system were the servos since they demanded the most amount of power. It was important to verify that the servo initial startup current would not exceed the 3 A limit of the circuit. The subsystem then focused on stepping down 6V to 3.3V to power the development kits. This circuitry was hooked up to the backup battery supply and switching circuitry along with the development board and servos. The power system was tested with the full load and tested that both the wall power and backup battery pack could handle the load.

The screen backlight required 12 V to be powered. This circuitry required extensive testing and redesigning to supply the screen with enough voltage to keep the screen fully illuminated while the servos were dispensing. While testing, the unloaded voltage was compared to the voltage with the screen connected. The first circuit designs could not handle the screen load and resulted in the voltage sagging. Eventually, the 12 V circuitry used in the final design was developed and tested successfully.

Once the PCBs were assembled and integrated into the product, the entire system was tested both on wall plug supply and the backup power supply. The system successfully ran on both the wall plug and battery pack. The device can also successfully switch between the two systems automatically.

Authors: RL

Pill Dispensing

The dispensing subsystem was tested by ensuring that the pills were picked from their containers correctly, detected in the feedback sensors and released from the gate with no pills getting stuck.

The system's dispensing reliability was evaluated by running dispenses multiple times with the container and disc. The disc was designed with the slots at different angles and depth. Multiple iterations of the disc were crucial to picking a design that maximized picked pills and minimized stoppages. The container had blockers on the inside to minimize jams as well. The blockers were printed at different heights from 1-10mm with increments of 1mm to decide which height allowed minimum stoppages.

To validate the detection system, the phototransistors' ability to register a pill dropping was tested in the funnel. Due to the inconsistency of lighting in each funnel, the code and build of the funnel had to have a working tolerance. The phototransistor expected to go high when a pill passed and remained low when nothing happened. The phototransistor and LED were placed on the top and bottom of the funnel, respectively.

The gate had to be big enough, so no pills slipped out once they dropped from the funnel and small enough to be able to fit other components in the system.

The entire dispensing process was tested by running full dispenses. Once a dispense was initiated, a good dispense looked like servos spinning the disc repeatedly in a pill picking cycle with no jams. The pills were expected to pass through the chute, be detected by the feedback sensors, and then be dispensed by opening the gate.

Authors: JP

Wi-Fi Connectivity and Logged Data Storage

Wi-Fi functionality was tested by connecting other devices to its access point and making sure that every HTTP message was received correctly and responds appropriately. The parts involved in making sure this subsystem operates correctly are receiving the HTTP response, parsing it, interfacing with the flash chip to retrieve the requested information, and building the response correctly.

Testing this submodule followed a two-step pattern:

1. Send an HTTP request to write/update information in the pill dispenser
2. Send an HTTP request to read the same information to ensure that it was updated correctly

This form of testing verifies all parts of the subsystem: HTTP message handling, and flash chip interfacing.

Authors: RO

User Interface

Test patterns were used to test the screen. These test patterns were real photos that were digitized into a bitmap that the program could read and then were displayed to the screen. This test procedure was used to determine if the screen had any visible tearing and to test signal integrity at various SPI clock speeds. Through this testing, an optimal

SPI bus speed was determined by visually inspecting the images for imperfections and increasing or decreasing the speed accordingly.

To test the user input, a small test program was created that displayed all user inputs to the screen as a rolling text output with each user input being given its own line. The buttons and encoder were then actuated. This created an output that should correspond with the order and quantity in which the buttons and encoder were actuated. Through this testing program, problems could be found in the buttons and encoder with switch bouncing.

To test the user interface state machine, the state machine was first mapped. After the state machine was mapped, a test could be devised to systematically test every connection in the state machine. By stepping through each connection in the state machine, any difference from the planned state machine map could be documented and fixed.

Authors: CB

6. Mechanical Sketch:

The outer shell of the device is drawn below. Following the marketing requirements, it's designed to be 18 inches tall and 12 inches wide. The holes on the outer shell include 2 holes on the side for the switch and power connection, which are on the side, and holes along the front for the screen, buttons, and dial.

Authors: JP

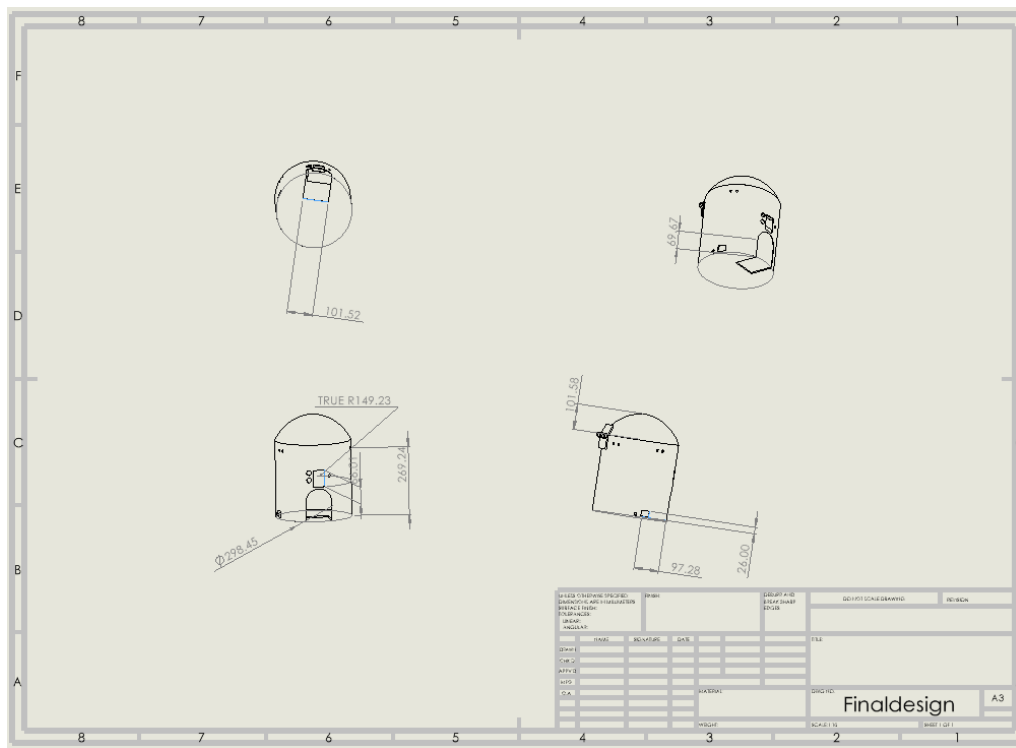


Figure 6.3.1 *Mechanical Sketch of Pill dispensing shell*

The bottom part of the shell is shown below. This sketch shows the mounting mechanism for the PCB components. The mounting mechanism for the power board PCB is two extruded cylinders, at 95mm high, with holes at the ends to fix the PCB at the bottom of the shell. There is a mounting platform along the front, at 219.8mm high, to mount the User interface components and the microcontroller/Wi-Fi boards.

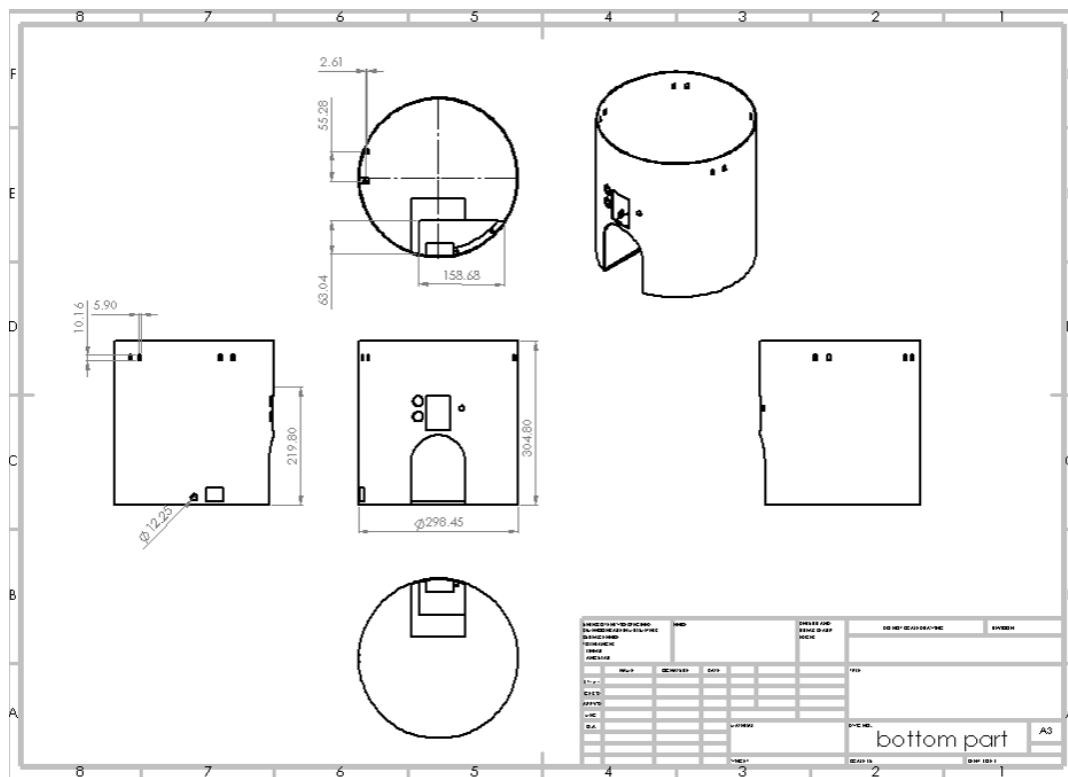


Figure 6.3.2 *Mechanical Sketch of bottom part of Pill dispensing shell*

The SolidWorks part is shown below to give a better idea of how the inner components are placed relative to each other.

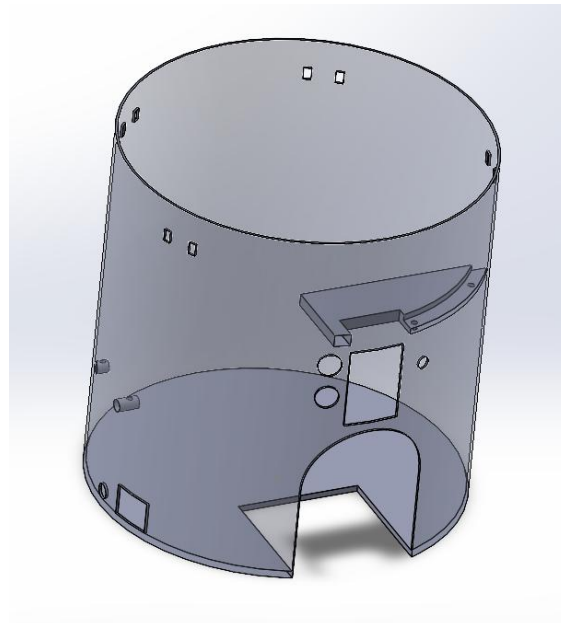


Figure 6.3.3 *SolidWorks Assembly of bottom part of Pill dispensing shell*

The disc below is the floor of the pill container, fitted with slot that will catch pills. The slots in the disc are sized with a tolerance of $\pm 1.5\text{mm}$ to ensure just one pill can fit in a slot. The disc has 3 slots to increase the likelihood of catching a pill, which will increase the dispensing speed of the device.

Each slot measures 7mm in width and 20mm in length with a filet of 5mm. The slots are angled to about 45 degrees to ensure minimal blockage and increase the chance that a pill falls into the slot. The disc is 100 mm in diameter due to the constraint of the pill dispenser being maximum 457mm. The extruded part in the middle is where the servo motor will be mounted; the slot on top is designed to fit the motor piece tightly to ensure accurate movement. Getting the disc to be as small as possible while still maintaining the security of one pill per slot is crucial to meet engineering and marketing requirements.

Authors: JP

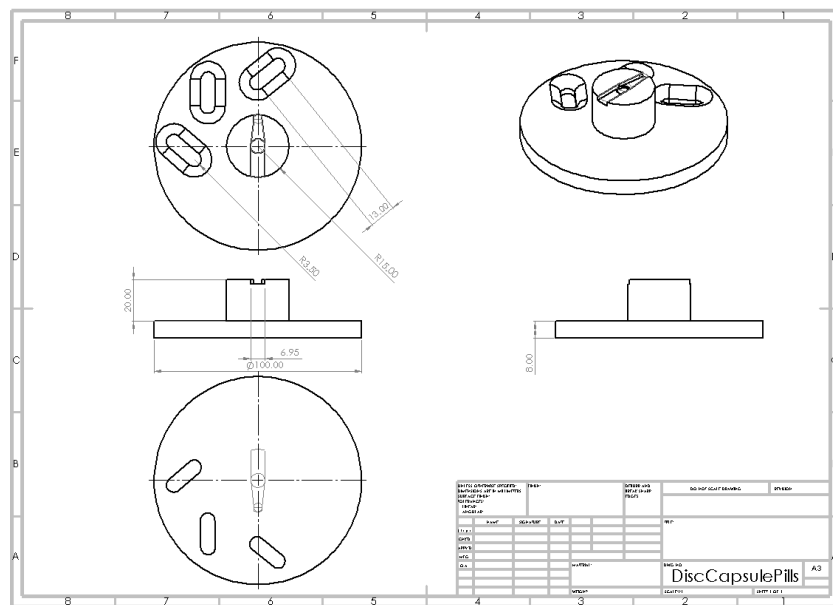


Figure 6.3.4 *Mechanical Sketch of Disc*

Below is the container that serves as the connection between the user input and the disc. The container is 85mm tall with a radius of 100mm. It is split in half to allow for space for the motor along with ensuring that the 270 servo motors can operate with the space given. The slot on the front of the container allows the user to change the disc depending on the type of pill that will be put in the container; capsule, circle, etc. The container will hang from the shell using the two prongs on the back side. Two stoppers

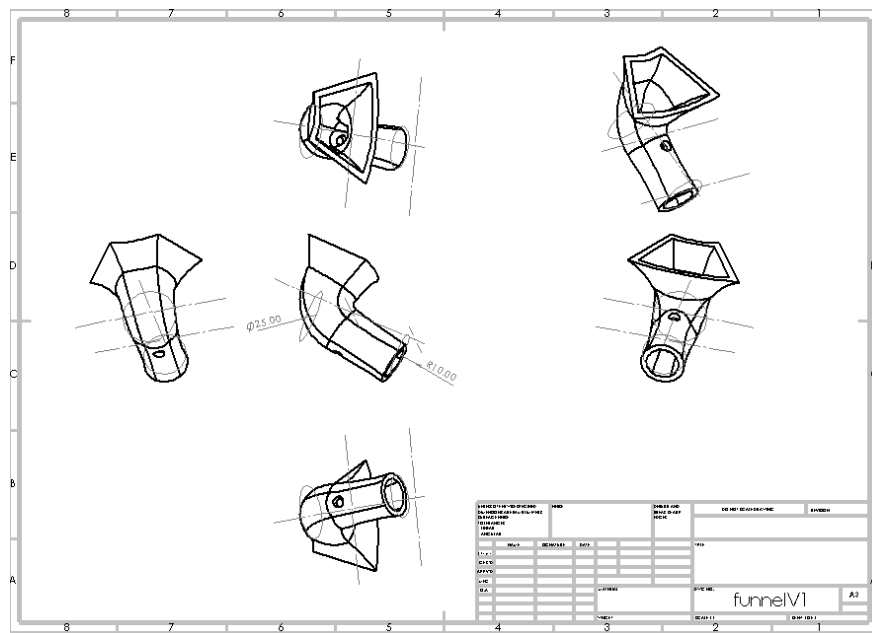


Figure 6.3.6 *Mechanical Sketch of Funnel*

Designing the motor mount required designing a precise and sturdy part that could be easily added to the already built container. The sketch below displays the shell which will hold the servo motor and the bracket which will be attached from the motor mount to the container.

Authors: JP

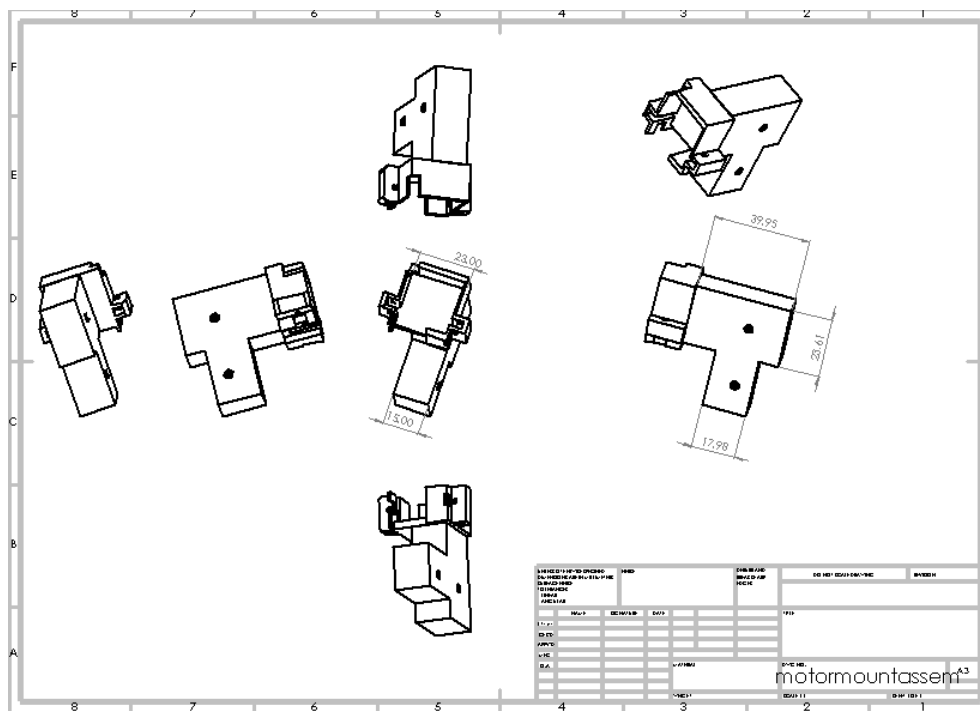


Figure 6.3.7 *Mechanical Sketch of Motor Mount*

Due to centrifugal force the disc slid out of its place and caused the motor to get stuck. To fix this issue, a C-shaped part with magnets at each end was installed to prevent the disc from sliding out. The disc needed to still be able to be removed to change the disc depending on what type of pill is in the container; this magnet makes it possible.

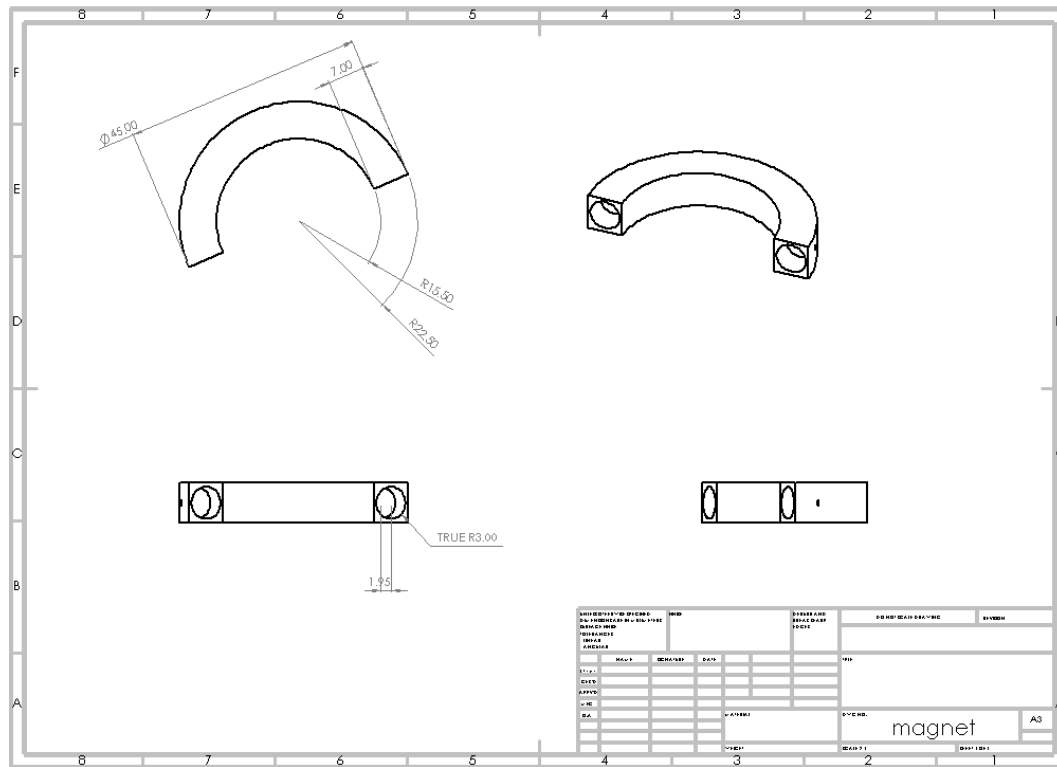


Figure 6.3.8 *Mechanical Sketch of Magnet*

After passing through the funnel, pills will consolidate at the gate, which is shown below. The gate consists of a shelled hole shaped product with a base that is 74mm high, 119mm wide, and 101.60mm in depth. The gate has a compartment for the motor that will rotate a gate door, which will be where the final dispense will occur. The gate has two legs that make it slide and fit along the base of the shell.

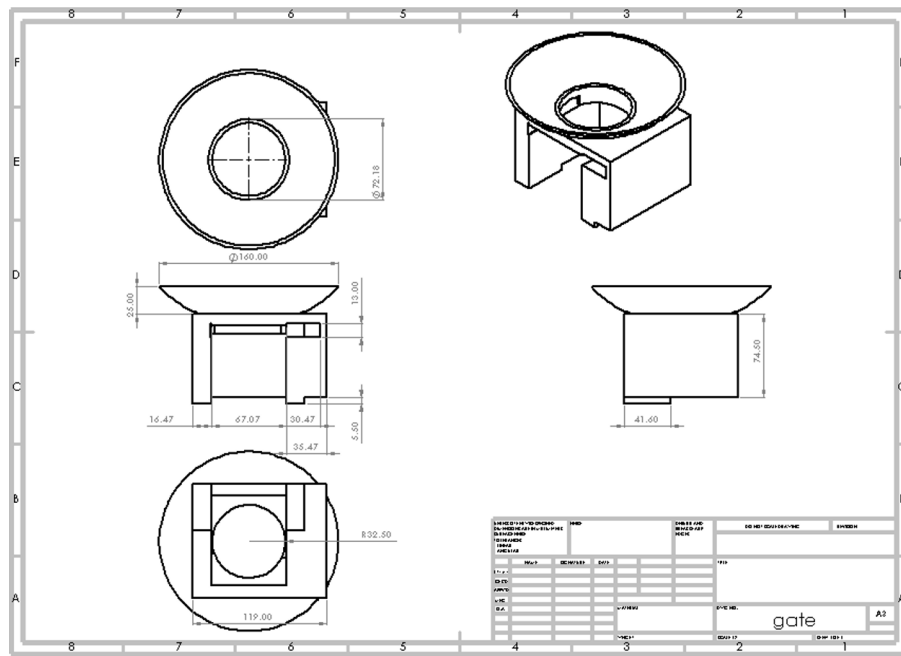


Figure 6.3.9 *Mechanical Sketch of Gate*

The gate door below has a slot on its side where the servo arm is inserted and will be rotating open and closed.

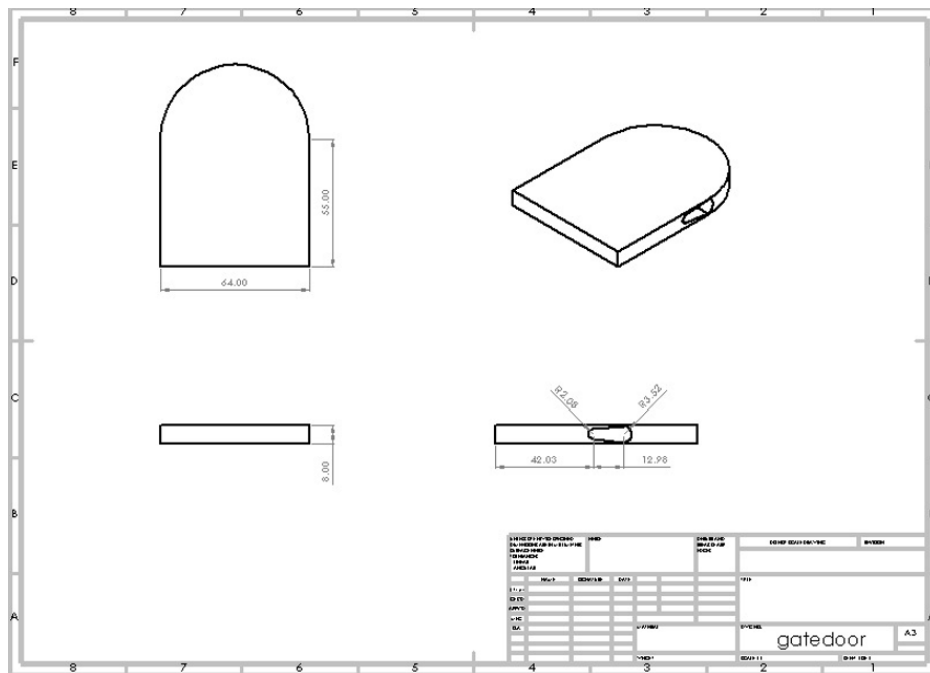


Figure 6.3.10 *Mechanical Sketch of Gate Door*

Below is a photograph of the dispensing system for the final demo. The container is mounted to the outer shell. The disk fits into the slot given by the container, and the

funnel is glued to the bottom slot of the container. The pills are filled to about $\frac{1}{4}$ - $\frac{1}{2}$ of the container and the disk is rotated by hand to mimic the rotation produced by a motor.

Authors: JP



Figure 6.3.11 *Mechanical Design Photograph*

7. Parts List

1.1. Parts list

Item #	BOM Level	Ref Des	Qty	PN	Description
1	PCB 1	C1	1	ECA-1CM100I	10uF Electrolytic Capacitor
2	PCB 1	C2	1	16ZLH1000MEFC10X16	1000uF Electrolytic Capacitor
3	PCB 1	C3	1	35ZLH680MEFC10X23	680uF Electrolytic Capacitor
4	PCB 1	C4	1	C323C102KAG5TA	1nF disk capacitor
5	PCB 1	C5,C6	2	M9619	22uF Electrolytic Capacitor
6	PCB 1	D1, D2, D3, D4, D5	5	1N4001	Diode 50 V 1A
7	PCB 1	K1	1	PE014005	General Purpose Relay SPDT (1 Form C) 5VDC Coil Through Hole
8	PCB 1	Q1,Q5,Q6,Q7,Q8,Q9	6	2N7000	N-Channel 60 V 200mA
9	PCB 1	R1,R4,R6,R7,R8,R10,R12,R14,R16,R17,R18,R19	12	CF14JT1K00	1k Ohm Resistor
10	PCB 1	R2,R5,R9,R11,R13,R15	6	CF14JT10K0	10k Ohm Resistor
11	PCB 1	R3	1	CFM12JT33K0	33k Ohm Resistor
12	PCB 1	U1	1	MC14069UBCPG	Invertor
13	PCB 1	U2	1	LM1117T-3.3	3.3 V Linear Regulator
14	PCB 1	U3	1	LM555xN	555 Timer/Oscillator 8-PDIP
15	PCB 2	U1	1	5369-BY25Q16AWTIG(R)TR-ND	IC FLASH 16MBIT SPI/QUAD 8SOP
16	Off Board	M1,M2,M3,M4,M5,M6	6	SER0046	9G 270deg Metal Servo 1.5kg
17	Off Board		2	NUCLEO-H533RE	MCU development board
18	Off Board		1	X-NUCLEO-67W61M1	WiFi development board
19	Off Board	SW1	1	PEC12R-4225F-N0024	Rotary Encoder with switch
20	Off Board	SW2	1	RP3502ABLK	Momentary SPST switch
21	Off Board	SW3	1	R5ABLKBLKFF0	SWITCH ROCKER DPST 20A 125V
22	Off Board		1	NHD-2.4-240320DA-CTXN	320x240 2.4 inch graphic display
23	Off Board		1	541324562	Molex 45 pin FCC connector .5 mm pitch
24	Off Board		1	FPC050P050	50 pin FCC connector breakout board
25	Off Board		1	4651	Fingerprint scanner
26	Off Board	Q1, Q2, Q3, Q4	4	QSD123	Phototransistors 880nm Top View Radial, 5mm Dia (T 1 3/4)
27	Off Board	D1, D2,D3, D4	4	HLMP-EG1A-Z10DD	Red 626nm LED Indication - Discrete 2.1V Radial

Table 7.1.1 *Parts List*

1.2. Materials Budget List

Qty.	Part Num.	Description	Unit	Total
			Cost	Cost
2	NUCLEO-H533RE	MCU development board	\$15.88	\$31.76
1	X-NUCLEO-67W8	WiFi development board	38.75	38.75
1	SER0043	SERVOMOTOR RC 4.8-6V	4.00	4.00
1	SER0008	SERVOMOTOR RC 4.8V	3.62	3.62
1	PEC12R-4225F-N	Rotary Encoder with switch	1.39	1.39
3	RP3502ABLK	Momentary SPST switch	2.51	7.53
1	NHD-2.4-240320C	320x240 2.4 inch graphic display	14.90	14.90
1	541324582	Molex 45 pin FCC connector .5 mm pitch	2.20	2.20
1	FPC050P050	50 pin FCC connector breakout board	7.89	7.89
10	QSD123	Phototransistors 880nm Top View Radial, 5mm Dia (T 1 3/4)	0.48	4.80
10	HLMP-EG1A-Z10I	Red 628nm LED Indication - Discrete 2.1V Radial	0.64	6.40
1	3859	Battery Holder (Open) AA 4 Cell Wire Leads - 6.5" (165.1mm)	2.95	2.95
1	PE014005	General Purpose Relay SPDT (1 Form C) 5VDC Coil Through	2.94	2.94
2	NUCLEO-H533RE	MCU development board	15.88	31.76
1	X-NUCLEO-67W8	WiFi development board	38.75	38.75
1	SER0043	SERVOMOTOR RC 4.8-6V	4.00	4.00
1	SER0008	SERVOMOTOR RC 4.8V	3.62	3.62
1	PEC12R-4225F-N	Rotary Encoder with switch	1.39	1.39
3	RP3502ABLK	Momentary SPST switch	2.51	7.53
1	NHD-2.4-240320C	320x240 2.4 inch graphic display	14.90	14.90
1	541324582	Molex 45 pin FCC connector .5 mm pitch	2.20	2.20
1	FPC050P050	50 pin FCC connector breakout board	7.89	7.89
10	QSD123	Phototransistors 880nm Top View Radial, 5mm Dia (T 1 3/4)	0.48	4.80
10	HLMP-EG1A-Z10I	Red 628nm LED Indication - Discrete 2.1V Radial	0.64	6.40
1	3859	Battery Holder (Open) AA 4 Cell Wire Leads - 6.5" (165.1mm)	2.95	2.95
1	PE014005	General Purpose Relay SPDT (1 Form C) 5VDC Coil Through	2.94	2.94
1	4651	Fingerprint scanner	39.95	39.95
4	5369-BY25Q16AV	IC FLASH 16MBIT SPI/QUAD 8SOP	0.23	0.92
2	LM1117T-3.3/NOF	IC REG LINEAR 3.3V 800MA TO220-3	1.61	3.22
1	R5ABLKBLKFF0	SWITCH ROCKER DPST 20A 125V	2.23	2.23
4	SER0048	9G 270DEG METAL SERVO 1.5KG	6.90	27.60
1	MC7808CT-BP	IC REG LINEAR 8V 1A TO220AB	1.27	1.27
2	SER0048	9G 270DEG METAL SERVO 1.5KG	6.90	13.80
1	8V 3A Power Supl	8V 3A Power Supply	10.98	10.98
1	TBA 1-0512	DC DC CONVERTER 12V 960MW	3.29	3.29
2	NUCLEO-H533RE	MCU development board	15.88	31.76
2	SN74HC14N	14 pin DIP schmitt trigger inverter	1.05	2.10
1	NUCLEO-H563ZI	NUCLEO-144 STM32H563ZI EVAL BRD	29.92	29.92
2	SER0048	9G 270DEG METAL SERVO 1.5KG	6.90	13.80
3	TBA 1-0512	DC DC CONVERTER 12V 960MW	3.29	9.87
10	BS170	N-Channel 80 V 500mA (Ta) 830mW (Ta) Through Hole TO-9	0.27	2.70
10	1N5400	DIODE STANDARD 50V 3A DO201	0.19	1.90
5	MF-R300	PTC RESET FUSE 30V 3A RADIAL	0.48	2.40
3	CD4069UBE	IC INVERTER 6CH 1-INP 14DIP	0.90	2.70
3	MC33063AP	IC REG BUCK BOOST ADJ 1.5A 8PDIP	1.45	4.35
10	1N5819	DIODE SCHOTTKY 40V 1A DO41	0.09	0.90
3	13R224C	FIXED IND 220UH 700MA 470 MOHM	0.59	1.77
1		12 Values 0803 SMD Capacitor Kit	9.99	9.99
25	RMCF0803FT10K	RES 10K OHM 1% 1/10W 0803	0.01	0.25
25	RMCF0803FT4K7	RES 4.7K OHM 1% 1/10W 0803	0.01	0.25
25	RMCF0803FT1K0	RES 1K OHM 1% 1/10W 0803	0.01	0.25
25	RMCF0803ZT0R0	RES 0 OHM JUMPER 1/10W 0803	0.01	0.25
25	RMCF0803FT100I	RES 0 OHM JUMPER 1/10W 0803	0.01	0.25
4	SN74HC14DR	IC INVERT SCHMITT 6CH 1IN 14SOIC	0.40	1.60
10	SML-D12U1WT86	LED RED DIFFUSED 1608 SMD	0.08	0.80
2	STM32H563ZIT6	IC MCU 32BIT 2MB FLASH 144LQFP	8.89	17.78
1	PRPC020DFBN-R	CONN HEADER VERT 40POS 2.54MM	1.13	1.13
3	OSTTE020104	TERM BLK 2POS SIDE ENT 3.5MM PCB	0.48	1.44
1	DKS-CR2032H	BATTERY HOLDER CR2032 6" LEADS	1.78	1.78
2	PE014005	General Purpose Relay SPDT (1 Form C) 5VDC Coil Through	4.03	8.06
2	MC33063AP	IC REG BUCK BOOST ADJ 1.5A 8PDIP	1.64	3.28
8	BS170	N-Channel 80 V 500mA (Ta) 830mW (Ta) Through Hole TO-9	0.43	2.58
10	1N5400	DIODE STANDARD 50V 3A DO201	0.35	3.50
2	MC7808CT-BP	IC REG LINEAR 8V 1A TO220AB	1.27	2.54
3	MF-R300	PTC RESET FUSE 30V 3A RADIAL	0.57	1.71
Total			\$523.13	

2. Table 7.2.1 Material Budget List

8. Lessons Learned

Analog to digital converter issues

The dispensing design of this project relied on an analog to digital converter for feedback to confirm that a pill dropped through a channel. This worked well on the development kits that were being used to originally test. Later in the project when it came time to test on the PCB, there was an issue with the ADC converter failing to read voltage values accurately. Triggering an ADC conversion led to a spike in voltage that always caused readings to be significantly higher than expected. This caused pill detection readings to be incorrect. The team solved this issue by using general purpose interrupt pins with a BJT inverting amplifier circuit. This circuit pulled the interrupt pins high to 3.3 V when a pill passed through, blocking the LED supplying light to the phototransistor within the chute. While this solution fixed the problem, it is not as robust as the original ADC design. The ADC was more capable of adjusting to rooms with varying amounts of light. The team made several conclusions based on this issue. The first conclusion was that the PCB should have been created much earlier in the semester. Ordering the PCB later in the semester meant that there was not much time to find issues, and design/order another. The second conclusion made was that exposing more pins than necessary in a prototype PCB is crucial. This allows for quick fixes to be made in case something goes wrong. In this project, four extra GPIO pins became crucial for completing the project on time.

Consult other engineering disciplines for expertise

The mechanical design was completed entirely by the team. While it worked very well with minimal jams, consulting a mechanical engineer would have been a better approach. Consulting other engineering disciplines could have provided ideas for creating a pill dispenser that's more compact, efficient, robust, and reliable.

9. Team Information

Connor Beaven, Computer Engineering and Computer Science

Rachelle LaBrie, Electrical Engineering

Ryan Oderkirk, Computer Engineering and Computer Science

Josue Panchana, Electrical Engineering

10.Project Schedules

Design Gantt Chart

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Add New Column
1		SDP1 Fall 2025	253 days?	Sun 9/7/25	Wed 8/26/26			
2		Update Material From Seminar	0 days	Wed 9/24/25	Wed 9/24/25			
3		Need Statement						
4		Objective						
5		Marketing Requirements						
6		Engineering Requirements	0 days?	Sun 9/7/25	Sun 9/7/25	2		
7		First Draft Due	1 day	Sun 9/7/25	Sun 9/7/25			
8		Complete Final Draft	1 day?	Sun 9/7/25	Sun 9/7/25			
9		Engineering Analysis	10 days	Wed 9/24/25	Tue 10/7/25	2,6		
10		General Design and Engineering Analysis	1 day	Wed 9/24/25	Wed 9/24/25			
11		TBD/TBR Analysis						
12		Functional Block Diagrams	13 days	Sun 9/14/25	Tue 9/30/25	2,6,10		
13		First Draft Due	1 day	Sun 9/14/25	Sun 9/14/25			
14		Complete Final Drawing						
15		Software Behavioral Models	1 day?	Sun 9/7/25	Sun 9/7/25			
16		Mechanical Sketch	51 days	Wed 9/24/25	Wed 12/3/25	12		
17		Update Existing Sketch	13 days	Thu 9/25/25	Sun 10/12/25			
18		Complete Final Version	17 days	Mon 10/13/25	Tue 11/4/25	17		
19		Engineering Standards	3 days	Sun 9/28/25	Tue 9/30/25	6,12		
20		First Draft Due (Safety standards)	2 days	Sun 9/28/25	Mon 9/29/25			
21		Power standards					Rachelle	
22		Complete Final Draft						
23		Electrical Design	43 days?	Sun 10/5/25	Tue 12/2/25	52		
24		Pick Motors	4 days	Wed 9/24/25	Sun 9/28/25		Josue	
25		PCB Model revision 1	40 days	Wed 9/24/25	Tue 11/18/25		Josue	
26		Power system circuit schemaitc	40 days	Wed 9/24/25	Tue 11/18/25		Josue,Rachelle	
27		Sensor sub circuits schematic	40 days	Wed 9/24/25	Tue 11/18/25		Josue,Rachelle	
28		Power circuit PCB	25 days	Fri 11/28/25	Thu 1/1/26		Rachelle	
29		Design MCU Board Circuit	15 days	Fri 12/12/25	Thu 1/1/26		Rachelle	
30		MCU PCB	27 days	Thu 1/1/26	Fri 2/6/26		Josue,Rachelle	
31		Design Mounting for Sensors	37 days	Thu 1/1/26	Fri 2/20/26		Josue,Rachelle	
32		Test PCBs	22 days	Fri 1/30/26	Sat 2/28/26		Josue,Rachelle	
33		<New Task>PCB Revisions	5 days	Mon 2/23/26	Fri 2/27/26		Josue,Rachelle	
34		Software Design	87 days	Sun 10/5/25	Sat 1/31/26	2,6,12		
35		Choose MCU	4 days	Wed 9/24/25	Sun 9/28/25		Ryan,Connor	
36		Choose Screen/UI	4 days	Wed 9/24/25	Sun 9/28/25		Connor,Ryan	
37		Choose input buttons and radial dial	4 days	Wed 9/24/25	Sun 9/28/25		Connor,Ryan	
38		Choose Wifi chip	7 days	Wed 10/1/25	Thu 10/9/25	19	Connor,Ryan	
39		RTOS research			Sun 10/12/25		Connor,Ryan	
40		Button and rotary dial development	13 days	Mon 10/13/25	Wed 10/29/25		Connor	
41		LCD Screen Development	13 days	Mon 10/13/25	Wed 10/29/25		Connor	
42		Motor Controller Development	13 days	Mon 10/13/25	Wed 10/29/25		Ryan	

		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Add New Column
43			Flash WiFi chip	44 days	Sun 11/2/25	Wed 12/31/25		Ryan	
44			WiFi API development	46 days	Sun 11/30/25	Fri 1/30/26		Ryan,Connor	
45			Flash Chip drivers						
46			Fingerprint scanner development	46 days	Sun 11/30/25	Fri 1/30/26		Ryan	
47			Mechanical Design	43 days?	Mon 10/6/25	Wed 12/3/25	6,9,12		
48			Pick filament			Sun 9/28/25			
49			CAD Model Revision 1	4 days	Wed 10/8/25	<u>Sun 10/12/25</u>	16	Josue	
50			CAD model revision 2	38 days	Mon 10/13/25	Wed 12/3/25	49	Josue	
51			<New Task>						
52			Midterm Report	0 days	Sun 10/5/25	Sun 10/5/25	16,19,6,9,12,15,8		
53			Prepare Midterm Report						
54			Midterm Report Due	1 day	Sun 10/5/25	Sun 10/5/25			
55			Subsystem Testing	48 days?	Wed 9/24/25	Fri 11/28/25			
56			Select Parts	4 days	Wed 9/24/25	Sun 9/28/25		Josue,Rachelle,Rye	
57			Order Parts	1 day	Sun 10/12/25	Sun 10/12/25		Connor,Josue,Rach	
58			Receive Parts	1 day	Mon 10/20/25	Mon 10/20/25			
59			Testing	1 day?	Tue 10/21/25	Tue 10/21/25	58		
60			Subsystem Demos	10 days	Mon 11/17/25	Fri 11/28/25	57		
61			<New Task>						
62			Posters	0 days	Sun 10/26/25	Sun 10/26/25	19,23,34,47,52,55		
63			Prepare Posters						

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Add New Column
52		Midterm Report	0 days	Sun 10/5/25	Sun 10/5/25	16,19,6,9,12,15,8		
53		Prepare Midterm Report						
54		Midterm Report Due	1 day	Sun 10/5/25	Sun 10/5/25			
55		Subsystem Testing	48 days?	Wed 9/24/25	Fri 11/28/25			
56		Select Parts	4 days	Wed 9/24/25	Sun 9/28/25		Josue,Rachelle,Rya	
57		Order Parts	1 day	Sun 10/12/25	Sun 10/12/25		Connor,Josue,Racl	
58		Receive Parts	1 day	Mon 10/20/25	Mon 10/20/25			
59		Testing	1 day?	Tue 10/21/25	Tue 10/21/25	58		
60		Subsystem Demos	10 days	Mon 11/17/25	Fri 11/28/25	57		
61		<New Task>						
62		Posters	0 days	Sun 10/26/25	Sun 10/26/25	19,23,34,47,52,55		
63		Prepare Posters						
64		Posters Due	1 day	Sun 10/26/25	Sun 10/26/25			
65		Oral Presentations	7 days	Tue 10/21/25	Wed 10/29/25	62	Connor,Josue,Rya	
66		Prepare						
67		Slides Due	1 day	Tue 10/21/25	Tue 10/21/25			
68		Presentations	6 days	Wed 10/22/25	Wed 10/29/25			
69		Final Report	11 days	Wed 11/19/25	Wed 12/3/25		Connor,Josue,Rya	
70		Final Parts List	3 days	Mon 12/1/25	Wed 12/3/25		Connor,Josue,Rya	
71		SDP2 Spring 2026 (you can keep all of this the same for now)	158 days?	Mon 1/19/26	Wed 8/26/26			
71		SDP2 Spring 2026 (you can keep all of this the same for now)	158 days?	Mon 1/19/26	Wed 8/26/26			
72		Order Parts	0 days?	Mon 1/19/26	Mon 1/19/26			
73		Mechanical Design	35 days?	Mon 1/19/26	Fri 3/6/26		Josue	
74		Enclosure	35 days	Mon 1/19/26	Fri 3/6/26		Josue	
75		Model	25 days	Mon 1/19/26	Fri 2/20/26		Josue	
76		Locking design	20 days	Mon 1/19/26	Fri 2/13/26		Josue	
77		Dispensing Design	10 days?	Mon 1/19/26	Fri 1/30/26		Josue	
78		Disc	10 days	Mon 1/19/26	Fri 1/30/26		Josue	
79		Container	10 days	Mon 1/19/26	Fri 1/30/26		Josue	
80		Funnel	10 days	Mon 1/19/26	Fri 1/30/26		Josue	
81		Gate design	25 days?	Mon 1/19/26	Fri 2/20/26		Josue	
82		Bowl	25 days	Mon 1/19/26	Fri 2/20/26		Josue	
83		Gate	25 days	Mon 1/19/26	Fri 2/20/26		Josue	
84		Electrical Design	69 days?	Mon 1/12/26	Thu 4/16/26		Rachelle,Josue	
85		Testing	45 days	Mon 1/12/26	Fri 3/13/26		Rachelle,Josue	
86		Full load Power Test	10 days	Mon 1/26/26	Fri 2/6/26		Rachelle	
87		Test dispense sensor circuit	25 days	Mon 2/9/26	Fri 3/13/26		Rachelle,Josue	
88		PCB Testing	10 days	Mon 3/2/26	Fri 3/13/26		Rachelle,Josue	
89		Power Board Design	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle	
90		Power Board Layout	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle	
91		12 V Implementation	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle	
92		MCU Board Layout	20 days?	Mon 1/12/26	Fri 2/6/26		Rachelle,Josue	
93		Research existing boards	7 days	Mon 1/19/26	Tue 1/27/26		Rachelle	
94		Determine Pin Layout	10 days	Mon 1/19/26	Fri 1/30/26		Rachelle,Josue,Coi	
95		Design supporting circuits	20 days?	Mon 1/12/26	Fri 2/6/26		Rachelle	
96		UI Button Debouncing	15 days	Mon 1/12/26	Fri 1/30/26		Rachelle,Connor	
97		Dispense Sensors	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle	
98		WiFi Module	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle,Ryan	
99		MIDTERM: Demonstrate Hardware Subsystems	1 day?	Tue 3/10/26	Tue 3/10/26			
100		Finger Print Sensor	20 days	Mon 1/12/26	Fri 2/6/26		Rachelle	
101		Hardware Implementation	34 days?	Mon 2/23/26	Thu 4/9/26		Rachelle,Josue	
102		Power System with Power Board	28 days	Mon 2/23/26	Wed 4/1/26		Rachelle,Josue	
103		Implement MCU PCB Into System	35 days	Mon 2/23/26	Fri 4/10/26		Rachelle,Josue	
104		Mount PCBs and other components	31 days	Mon 3/2/26	Mon 4/13/26		Rachelle,Josue	
105		Software Design	112 days?	Fri 11/28/25	Mon 5/4/26		Connor,Josue,Rya	
106		Develop Software	61 days?	Mon 1/12/26	Mon 4/6/26		Connor,Josue,Rya	
107		UI	22 days?	Thu 1/15/26	Fri 2/13/26		Connor	
108		Develop State Machine	1 day	Thu 1/15/26	Thu 1/15/26		Connor	
109		Create Menus	12 days	Thu 1/15/26	Fri 1/30/26		Connor	
110		Create Keyboard	5 days	Mon 2/2/26	Fri 2/6/26		Connor	
111		Test and Debug	22 days	Thu 1/15/26	Fri 2/13/26		Connor	
112		Data logging	10 days?	Mon 1/26/26	Fri 2/6/26		Ryan	

113			Interface with Flash Chip	5 days	Mon 1/26/26	Fri 1/30/26		Ryan	
114			Design Data Format	5 days	Mon 1/26/26	Fri 1/30/26		Ryan	
115			Implement File System	5 days	Mon 2/2/26	Fri 2/6/26		Ryan	
116			Make Header Library	5 days	Mon 2/2/26	Fri 2/6/26		Ryan	
117			Test and Debug	10 days	Mon 1/26/26	Fri 2/6/26		Ryan	
118			✦ Security Pin	10 days?	Mon 2/16/26	Fri 2/27/26		Connor	
119			Interface with UI	5 days	Mon 2/16/26	Fri 2/20/26		Connor	
120			Make Library	10 days	Mon 2/16/26	Fri 2/27/26		Connor	
121			Test and Debug	10 days	Mon 2/16/26	Fri 2/27/26		Connor	
122			✦ Dispense Action	35 days?	Mon 1/12/26	Fri 2/27/26		Josue, Ryan	
123			Interface with ADC peripheral	5 days	Mon 1/26/26	Fri 1/30/26		Josue	
124			Create Dispense Sequence	5 days	Mon 2/2/26	Fri 2/6/26		Josue	
125			Make Library	5 days	Mon 2/9/26	Fri 2/13/26		Josue	
126			✦ WIFI	40 days?	Mon 1/12/26	Fri 3/6/26		Ryan	
127			Implement Sample Get Request	10 days	Mon 1/19/26	Fri 1/30/26		Ryan	
128			Format Data in JSON	5 days	Mon 2/16/26	Fri 2/20/26		Ryan	
129			Integrate JSON with logged Data	5 days	Mon 2/23/26	Fri 2/27/26		Ryan	
130			Integrate JSON into Get Request	5 days	Mon 3/2/26	Fri 3/6/26		Ryan	
131			✦ Real Time Clock	10 days?	Mon 3/2/26	Fri 3/13/26		Ryan, Connor	
132			Setup RTC	5 days	Mon 3/2/26	Fri 3/6/26		Connor	
133			Get interrupts from RTC	5 days	Mon 3/2/26	Fri 3/6/26		Ryan	
134			Handle interrupts from RTC	5 days	Mon 3/9/26	Fri 3/13/26		Ryan	
135			MIDTERM: Demonstrate Software Subsystems	1 day?	Tue 3/10/26	Tue 3/10/26			
136			RTOS Integration	5 days	Mon 3/23/26	Fri 3/27/26			
137			SUBMIT HW DRAFT HW SCHEMATICS	1 day	Sun 3/15/26	Sun 3/15/26			
138			✦ System Integration	26 days	Wed 3/11/26	Wed 4/15/26			
139			✦ Assemble Complete System	16 days	Mon 3/23/26	Mon 4/13/26	137,156		
140			Dispense Driver	14 days	Tue 3/24/26	Sun 4/12/26		Josue, Ryan	
141			Power System	14 days	Tue 3/24/26	Sun 4/12/26		Rachelle	
142			User Interface	14 days	Tue 3/24/26	Sun 4/12/26		Connor	
143			WiFi Module	14 days	Tue 3/24/26	Sun 4/12/26		Connor, Ryan	
144			✦ Test Complete System Integration	10 days	Wed 4/1/26	Tue 4/14/26	139		
145			Josue, Ryan	1 day	Tue 4/14/26	Tue 4/14/26	139	Josue, Ryan	
146			Power System	1 day	Tue 4/14/26	Tue 4/14/26	139	Josue, Rachelle	
147			User Interface	1 day	Tue 4/14/26	Tue 4/14/26	139	Connor, Ryan	
148			WiFi Module	1 day	Tue 4/14/26	Tue 4/14/26	139	Ryan	
149			✦ Revise Complete System Integration	11 days	Wed 4/1/26	Wed 4/15/26	144		
150			Dispense Driver	11 days	Wed 4/1/26	Wed 4/15/26		Josue, Ryan	
151			Power System	11 days	Wed 4/1/26	Wed 4/15/26		Josue, Rachelle	
152			User Interface	11 days	Wed 4/1/26	Wed 4/15/26		Connor, Ryan	
153			WiFi Module	14 days	Fri 3/27/26	Wed 4/15/26		Ryan	
154			Preliminary Demonstration of Complete System	1 day	Tue 4/14/26	Tue 4/14/26			
155			DEMO DAY: Project Demonstration and Presenta	1 day	Thu 4/16/26	Thu 4/16/26			
156			DEMO DAY: Project Demonstration and Presenta	1 day	Thu 4/16/26	Thu 4/16/26			
157			SUBMIT FINAL HW SCHEMATICS	1 day	Tue 4/21/26	Tue 4/21/26			
158			SUBMIT FINAL SW CODE	1 day	Tue 4/21/26	Tue 4/21/26			
159			✦ Develop Final Report	80 days	Tue 1/13/26	Sun 5/3/26			
160			Write Final Report	111 days	Fri 11/28/25	Sun 5/3/26			
161			Submit Final Report	1 day	Mon 5/4/26	Mon 5/4/26			

Table 10.1.1 Gantt Chart

10. Conclusions and Recommendations

Design Implementation

This project aims to develop a system to dispense pills for a client at pre-configured times. The system is a lower cost alternative to current options on the market. Most of the system will be 3D printable to make it as openly distributable as possible. While 3D printing is not entirely accessible, this project can still be a more affordable option and manufactured at a lower cost compared to other devices on the market. The device successfully operates as a smart pill dispenser to assist people in dispensing their medication while tracking its usage through logs. It can be connected to other common household devices through Wi-Fi to configure doses, set pill counts, and access usage information.

Team Dynamic

The completion of this project required various engineering skills. Mechanical knowledge was required to design the pill picking logic and model the parts. Electrical knowledge was required to design the circuitry, PCB, and power supply. Computer software skills were required to write logic to interact with all hardware and coordinate all features concurrently in one application. While no team member was restricted to one area: Josue mainly worked on the mechanical design, Rachelle focused on circuit and PCB design, Connor focused on the user interface, and Ryan focused on Wi-Fi connectivity, flash chip integration, pill dispensing logic, and FreeRTOS integration. Having teammates with different strengths helped allow the team to divide project work effectively and in parallel. The team worked to complete their subsystems independently first. As a subsystem was proven successful, it was then integrated with other functioning subsystems.

Author: RO

Future of the Project

Other advanced functionalities that could be added to this project include support for liquid medications and filling travel pill boxes. This project is designed with future expansion in mind, allowing for increasingly advanced features to be added over time. Overall, this project highlights how thoughtful engineering can be used to increase the quality of life of people with prescription medications.

Recommendations for future students

After three semesters of planning, designing, and implementing a fully functional smart pill dispenser, the team has learned many lessons that would have been helpful to know from the start.

One big recommendation is to understand the software a company supplies with the product in addition to the hardware. For example, the team opted to buy a Wi-Fi coprocessor to handle Wi-Fi related operations. After receiving the part, it was discovered that the device had to be flashed with software that didn't come pre-installed. This caused a two-week setback in the project because a separate program had to be written to program the Wi-Fi coprocessor. When choosing hardware, always understand how the manufacturer configures it to be used.

Another big recommendation is to order PCBs as soon as possible to begin testing. The team discovered an issue with the way the pill feedback logic worked on the custom PCB, 1 week before demo day. There was no time to order a new board, and some last-minute circuits had to be modified to get the feedback sensors working. While the project ended up working, much stress and headache could have been avoided if the team had ordered and worked on the PCB sooner.

It is always a good idea to order extra components. There were multiple occasions over the semester where a part got fried, misplaced, or broken. Having no backup parts could cause a multiple day set back.

If the team were to start over, they would prepare themselves in the software they would use to develop hardware/firmware. In this project, the team developed the firmware using ST's STMCubeIDE and PCB development using KiCad. There can be a steep learning curve for these tools, and it's important to start learning about them as soon as possible to make progress quickly.

While working on separate parts of the project is extremely effective, each team member must keep in mind that all parts will need to be integrated together. The team worked independently in the first semester on separate sub modules of the project. Going into the second semester, it became difficult to start integrating all the pieces together. From a software perspective, this meant combining programs for the screen and buttons, pills dispensing, Wi-Fi connectivity, and filesystem operations into one cohesive program. The team underestimated the time and effort required to put all this functionality together and highly recommends integrating throughout the semester rather than all at once.

11. References

- Abdul Minaam, D. S., & Abd-ELfattah, M. (2018). Smart drugs: Improving healthcare using smart pill box for medicine reminder and monitoring system. *Future Computing and Informatics Journal*, 3(4), 443–456.
<https://doi.org/10.1016/j.fcij.2018.11.003>
- Baarman, D. W., Eurich, S. T., Mollema, S. A., Dean, C. D., KUYVENHOVEN, N. W., Runyon, M. K., Van Den Brink, J. C., & Schamper, R. D. (2015). Pill dispenser.
- Clark, Amie. “Comparison: Automated Medication Dispensers.” *The Senior List*, 10 Dec. 2024, www.theseniorlist.com/medication/dispensers/.
- Cohen, R. A., & Mykyta, L. (2021). *Percentage of adults aged ≥ 18 years who took prescription medication during the past 12 months, by sex and age group — National Health Interview Survey, United States, 2021*. National Center for Health Statistics. <https://www.cdc.gov/nchs/nhis.htm>
- Eziya, STM_HAL_ILI9431. (2019). Github Repository.
https://github.com/eziya/STM32_HAL_ILI9341
- Genelin, M. P., Helmkamp, L. J., Steiner, J. F., Maertens, J. A., Hanratty, R., Vupputuri, S., Havranek, E. P., Dickinson, L. M., Blair, I. V., & Daugherty, S. L. (2023). Patient Pill Organization Strategies and Adherence Measured in a Cross-Sectional Study of Hypertension. *Patient preference and adherence*, 17, 817–826.
<https://doi.org/10.2147/PPA.S399693>
- Hartung, N. (2022, December). Technology to help you remember to take medications: Automatic pill dispensers. American Association of Psychiatric Pharmacists. Retrieved March 13, 2025, from <https://aapp.org/556051>
- Poirier, M., & Bouchard, P.-A. (2022, November 1). Portable medication dispenser.
- Savvides, M., Kyriacou, C., Kassianides, V., Markides, C., & Tatas, K. (2024). PDU: A Pill Dispenser Unit for an IoT System for Monitoring and Controlling the Pharmaceutical Treatment for Patients With Psychological Mental Conditions. In 2024 International Conference on Computer, Control, Informatics and its Applications (IC3INA) (pp. 111-116).

Vaish, A. (2024). Automated pill dispenser to improve medication compliance, efficiency, and accuracy. In 2024 IEEE World Forum on Public Safety Technology (WFPST) (pp. 192-196).